

DOCUMENTATION

OF

MICROSOFT COPILOT STUDIO AGENT

(ADOPTION VIRTUAL ASSISTANT)

Microsoft Copilot Studio Agent:

Microsoft Copilot Studio extends the capabilities of Copilot by allowing organizations to build and customize AI-powered copilots tailored to their needs. Copilot Studio is a graphical, low-code tool for building agents and agent flows (iaanw, 2023). Copilot Studio's ability to link to various data sources using prebuilt or bespoke connectors is one of its best features. Because of this versatility, users can develop and coordinate complex logic, guaranteeing that their agent experiences are both potent and user-friendly. Also, because of the platform's low-code interface, even non-technical users can access the power of AI.

A Microsoft Copilot Studio agent is a powerful AI companion that can manage a variety of interactions and duties, from figuring out the optimal course of action based on its instructions and context to addressing problems requiring intricate talks. To achieve your intended objectives, agents coordinate a variety of language models, instructions, context, knowledge sources, topics, tools, inputs, and triggers, (iaanw, 2023). Agents can interact with clients and staff in a variety of languages via websites, mobile applications, Facebook, Microsoft Teams, and any other channel that the Azure Bot Service supports. In this project, users can only access the agent via Microsoft 365 Copilot or Microsoft Teams only due to company's restrictions. By doing activities as part of a conversation or in response to a trigger to support users and organizations, they can also increase productivity.

Copilot Studio agents utilize AI capabilities and a unique Natural Language Understanding (NLU). NLU model to comprehend user input and provide the most appropriate subject in response. A subject is a section of an agent-user discussion thread. See Create and edit topics for further details. For example, a company wants to create an agent that can cater to any customer's questions. In the agent, a topic can be made and named as "Company's Basic Info". When someone ask question "Who is the founder of the company?", the agent uses natural language understanding (NLU) to understand the *intent* behind the question. The agent matches that intent to the best topic, the Company's Basic Info topic.

Nevertheless, the agent is unable to foresee every kind of query that users may pose. Copilot Studio has strong AI-powered features that make use of the most recent developments in NLU models to assist lessen this. The agent can automatically produce responses when it is connected to knowledge sources (iaanw, 2023). These answers don't rely on the agent maker to come up with themes for every scenario; instead, they are conversational and use simple language. Additionally, Copilot Studio generates topics from a brief description of your needs

using AI driven by the Azure OpenAI GPT model, which is also utilized in Bing (iaanw, 2023). In a same vein, you can edit and change any topic in your agent by explaining the modifications you wish to make.

5.0 Methodology

In order to design, develop, and implement an intelligent Copilot Studio agent that can facilitate Digital Technology Adoption throughout PETRONAS Upstream, this project used an organized development methodology. Platform selection, project ideation, training and workshops, development activities, system design, testing, and verification are the main phases of the process.

5.1 Microsoft Copilot Studio as the Development Platform

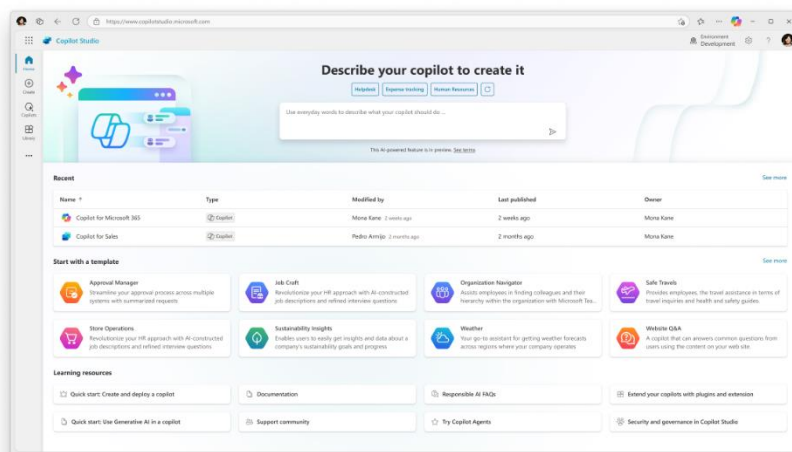


Figure 1.0 : Microsoft Copilot Studio website

Source : [[Microsoft Copilot Studio Website](https://www.copilotstudio.microsoft.com)]

As Microsoft Copilot Studio can create conversational AI agents linked to enterprise data through Microsoft Graph and the Microsoft 365 ecosystem, it was chosen as the primary platform. This integration guarantees that users receive precise and context-aware responses by enabling agents to access and interpret data from programs like Word, Excel, PowerPoint, and SharePoint. Furthermore, Copilot Studio facilitates the creation of parent and child agents for modular and topic-specific interactions, which improves scalability and maintainability.

Copilot Studio's ability in managing data and its strong conversational AI capabilities are two more important advantages. In order to keep information up to date, the platform can link to both dynamic sources for real-time changes and static documents like PDFs and Excel sheets. The agent can successfully analyze user questions, retrieve pertinent information, and deliver

guided responses that enhance the user experience because of its sophisticated prompt engineering skills. All of these characteristics make Copilot Studio the best option for creating an intelligent, scalable, and dependable adoption assistant that is customized to meet organizational requirements.

For this project, a Parent Agent which is AVA was created to oversee the entire Digital Technology Adoption process, while six Child Agents were designed to manage specific topic areas, ensuring accurate and detailed responses.

5.2 Project Activities

5.2.1 Workshops and Training

To equip the development team with the necessary skills to build a functioning Copilot agent, two Microsoft Copilot Studio workshops were conducted. These workshops were crucial in ensuring that the team comprehended Copilot Studio's technical capabilities and limitations, as well as how to organize agents, process data, and apply topic-specific recommendations for the adoption of digital technology.

Workshop 1: MS-4022: Extend Microsoft 365 Copilot in Copilot Studio



Figure 2.0 : MS-4022: Extend Microsoft 365 Copilot in Copilot Studio Workshop

Date: 12 September 2025

Objective: To provide a comprehensive understanding of Microsoft Copilot Studio, including its features, use cases, and practical applications for enterprise digital adoption.

Details:

- Carried out by Microsoft employees with expertise in Copilot Studio deployment.
- Discussed Microsoft Copilot Studio's architecture and essential capabilities, including how it integrates with Microsoft 365 applications like Word, Excel, PowerPoint, SharePoint, and Teams.
- Outlined how to expand Copilot Studio's features for situations unique to enterprises, with a focus on:
 - data intake from both live and static sources, such as PowerPoint, Excel, and PDFs.
 - Quick engineering for precise, context-aware answers.
 - To guarantee modular and scalable agent design, control the hierarchy of topics and agents.
- Demonstrated the practical applications of extending Copilot Studio, highlighting the importance of aligning agent capabilities with organizational objectives.
- Included practical activities in which participants investigated:
 - Building sample agent responses.
 - Connecting agents to data sources.
 - Testing queries for accuracy.

Outcome :

- Members of the team developed a solid grasp of Copilot Studio's capabilities and limitations.
- Explained the best way to organize Parent and Child Agents.
- Laid the groundwork for the upcoming workshop's practical development tasks.

Workshop 2: PL-7008 - Create Agents in Microsoft Copilot Studio

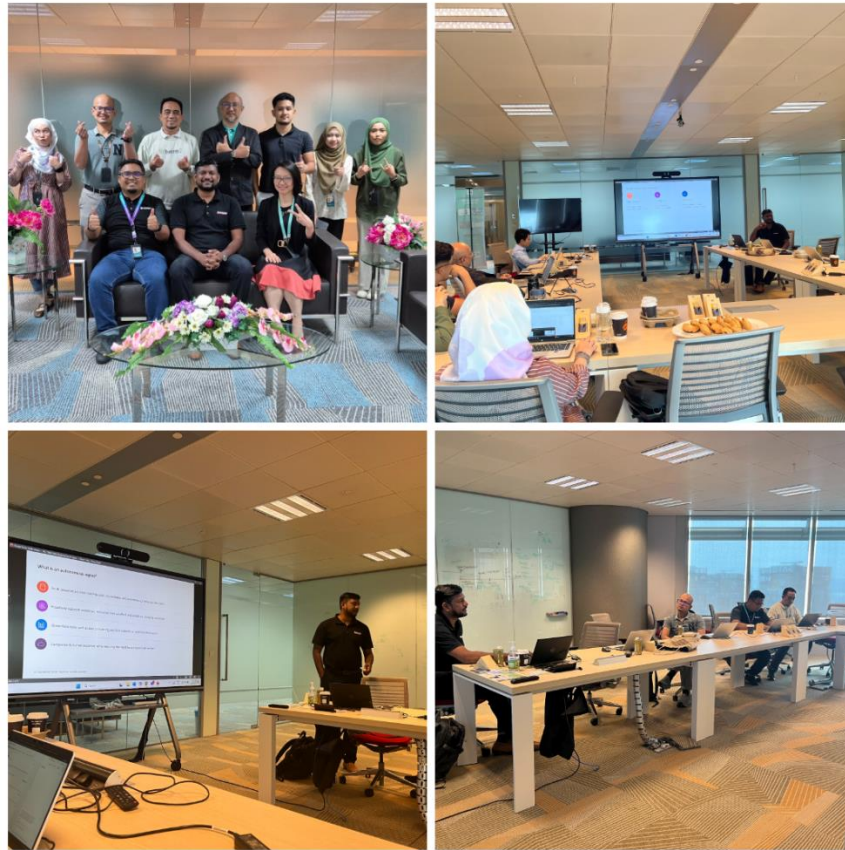


Figure 3.0 : PL-7008 - Create Agents in Microsoft Copilot Studio

Date: 22 September 2025

Objective: To provide hands-on training in creating and configuring Parent and Child Agents within Microsoft Copilot Studio.

Details:

- Focused on the implementation of hierarchical decision flow, subject management, and the practical development of agents.
- In order to apply training principles in the real world, team members learnt how to create their own solution and development environment within Copilot Studio.
- Among the activities were:
- Creating Parent and Child Agents specifically for subjects related to the use of digital technology.
- Setting up conditional prompts and subject flows for various user maturity levels (Beginner, Intermediate, Matured).

- Establishing connections between agents and actual static and dynamic data sources to guarantee precise and context-specific replies.
- Using example inquiries to test agent interactions and iteratively improving answers.
- Establishing a development environment that mimics actual business situations and enables complete solution deployment within Microsoft 365 Copilot.
- Utilizing optimization techniques and debugging methods to improve agent performance.

Outcome:

- Gained actual expertise in environment configuration, agent deployment, and solution design, bridging the gap between theoretical understanding and real-world implementation.
- The team was ready for iterative testing and AGILE-style development during the project phase.

5.3 System Development

5.3.1 Development Approach

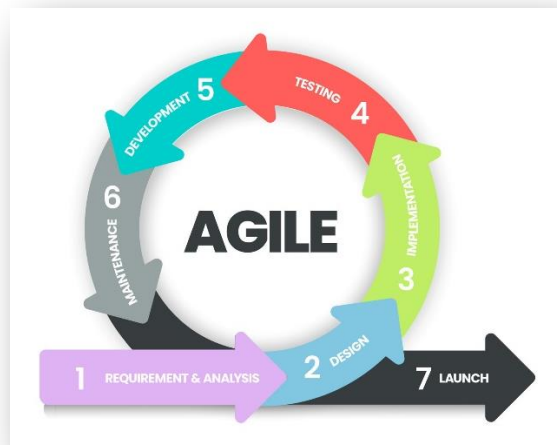


Figure 4.0 Agile Cycles in Software Development

The project followed an AGILE-style development methodology. Agile methodology is a framework for project management that divides projects into multiple dynamic periods, or sprints (Laoyan, 2025). The methodology used in the Agile framework is iterative. Teams

evaluate their performance at the conclusion of each sprint to determine what could be improved so they can modify their approach for the following one.

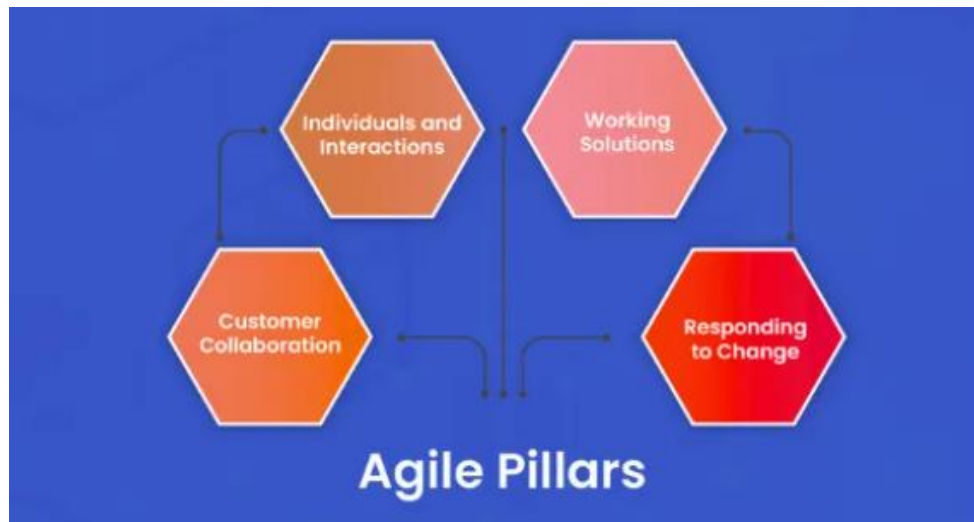


Figure 5.0 :Agile Pillars

The goals of this project are well aligned with the agile methodology. Agile's four pillars are individuals and interactions over processes and tools, working software over thorough documentation, customer collaboration over contract negotiation, and responding to change over following a plan (*What Is Agile Methodology? Types, Steps, Tools, Examples*, n.d.). These serve as a framework for the development process. Instead of concentrating only on documentation, the team made sure AVA supplied useful functionality by emphasizing user engagement and iterative feedback. The solution was able to swiftly adjust to changing adoption needs thanks to ongoing cooperation with team members and individuals in other departments, and integration with dynamic data sources was made possible by design flexibility. In keeping with the fundamental ideas of providing value in a dynamic environment, this Agile-driven strategy expedited development while guaranteeing that AVA remained user-centric, scalable, and adaptable to change.

5.3.2 Agent Design and Functionalities

AVA was designed to cater to different levels of user experience :

User Categorisation :

- **Beginner Users** : Step-by-step instructions with minimal prior knowledge required
- **Intermediate Users** : Provide detailed workflow of Adoption processes based on Digital & Technology Adoption Playbook as its main source of knowledge.
- **Matured Users**: Include strategic and predictive insights.

Functionalities

The agent provides guidance across multiple adoption tools. These tools are initiatives came up by the Adoption team to simplify the process of work done by the team and external party. Based on Digital & Technology Adoption Playbook (DTAP), the Adoption tools are :

1. Adoption Tools

- Adoption Target (AT)
- Voice of Customer (VoC)
- Adoption Maturity Evaluation (AME)
- Value Creation Plan (VCP)
- Adoption Change Readiness Assessment (ACRA)
- Capability & Change Management Plan (CCP)

2. Process Guidance

- Digital Technology Adoption Playbook (DTAP) integration
- Detailed instructions and information for adoption-related tasks
- Customized recommendations according on the user's level of maturity

5.4 System Structure & Design

- Agents: 1 Parent Agent (AVA) + 6 Child Agents

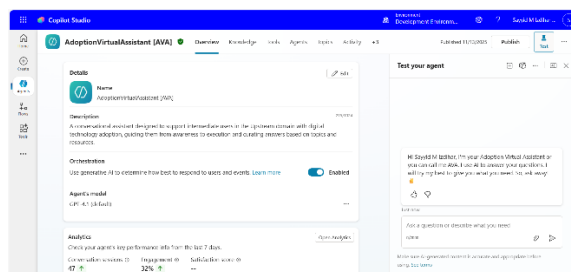


Figure 6.0 : Parent Agent (AVA)

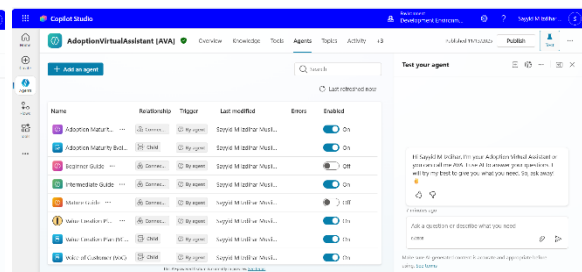


Figure 6.1 : 6 Child Agents

- Topics Covered: 33 distinct topics

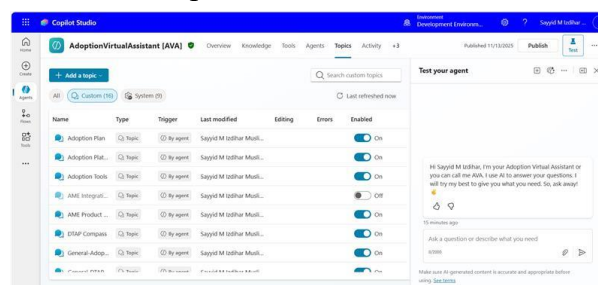


Figure 6.2 : List of topics

- Documents Connected: 68 PDFs, Excel, and PowerPoint files (static and live)

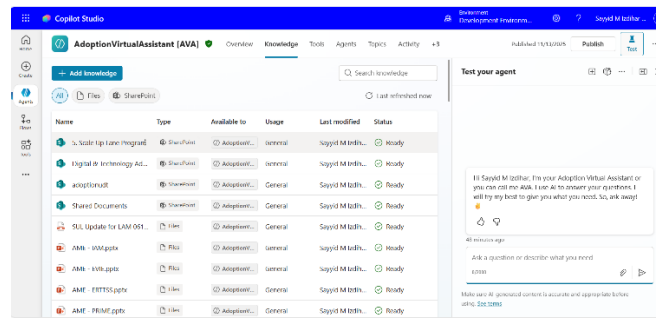


Figure 6.3 : List of knowledge sources

The system design adopts a hierarchical decision flow to ensure accurate and efficient guidance for users. At the top level, the Parent Agent which is AVA acts as the primary interface, handling general queries and determining the appropriate routing based on user intent. The Parent Agent is designed to cover several topics that correspond to the six adoption tools and acts as the primary point of contact for all user inquiries. Specialized topics which are General-DTAP, which answers inquiries about the Digital & Technology Adoption Playbook (DTAP) framework, and General-Adoption, which offers information on general adoption procedures, are included in this hierarchy. Every topic is arranged in a tree structure, starting with generative answers which is AI-generated responses that are intended to decipher user intent and offer dynamic, conversational solutions instead of static, pre-written text. Large language models are used in generative responses to produce contextually appropriate responses, guaranteeing adaptability and flexibility in user interactions. These generative responses are linked to certain knowledge sources, including curated documents and validated content repositories, in order to preserve accuracy and relevance. This method lowers the possibility of false information and increases user trust by ensuring that, even when responses are generated dynamically, they are based on reliable information.

When a query requires specialized knowledge, the Parent Agent delegates the request to one of the Child Agents, which are designed to manage topic-specific inquiries and deliver detailed, context-aware responses. This modular structure enhances scalability and maintainability, allowing updates to individual Child Agents without disrupting the overall system. The child agents embedded in AVA are one Intermediate Guide Agent, two Adoption Maturity Evaluation (AME) Agent, two Value Creation Plan (VCP) Agent, and Voice of Customer (VoC) Agent. Within Microsoft Copilot Studio, there are two

main ways to link child agents. Using the built-in features of Copilot Studio, the first approach is creating the child agent from scratch and then connecting it to the parent agent. The second approach, which prioritizes prompt engineering and guarantees smooth integration, builds a child agent straight from within the parent agent. The child agent's has access to the parent agent's knowledge sources, which ensures accuracy and consistency in responses, is a major benefit of the second strategy. During development phase, we first employed the first approach , however we ran into instability: the agents provided appropriate responses when tested as stand-alone agents, but returns inconsistent responses when accessed through the parent agent. We hypothesized that this resulted from the parent agent's inability to obtain information from the child agent. The problem was fixed and the agents provided the anticipated responses when the second approach was put into practice. However, in order to increase overall accuracy and stability, guarantee redundancy, and enhance system performance, we also kept the original agents developed using the first approach.

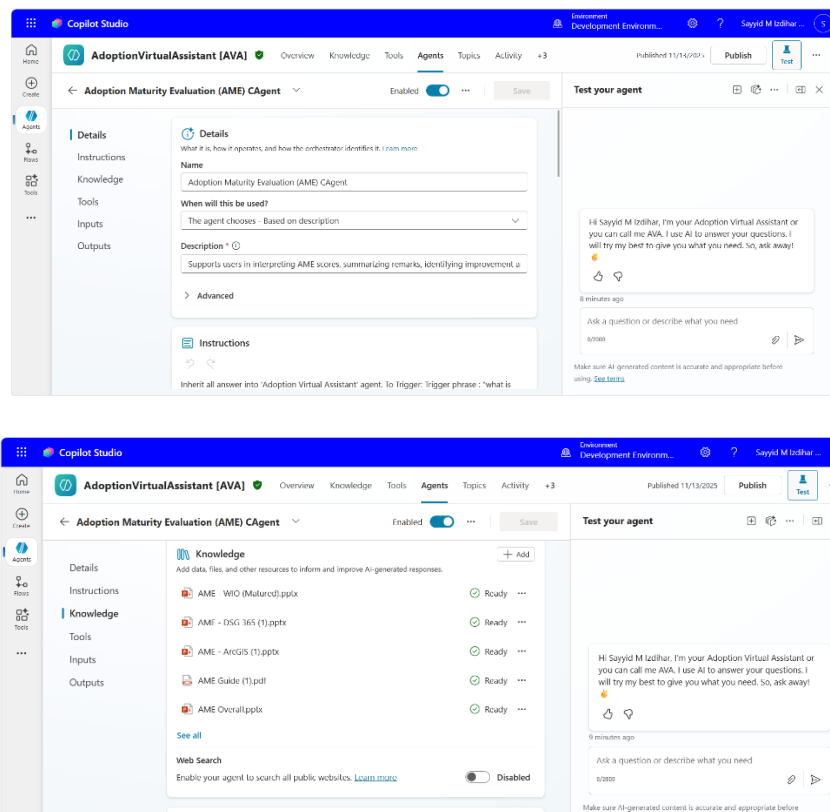


Figure 6.4 : Diagram showing the backend of Adoption Maturity Evaluation (AME) child agent built from inside the parent agent. The prompting is written on the instructions part. Any knowledge sources added will be stored directly inside parent agent.

5.5 Gantt Chart and Key Milestones

5.5.1 Gantt Chart

Task Name	July	August				September				October				November				December
	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1
PLANNING																		
Project assignment																		
Problem identification																		
Setting the objective for the project																		
Defining the scope of project																		
Finalise the project idea																		
Do further research on Microsoft Copilot Studio																		
Submission of project title																		
DEVELOPMENT																		
Beginner agent development																		
Intermediate agent development																		
Matured agent development																		
Testing and feedback gathering																		
Remedial																		
Workshop																		
DEPLOYMENT																		
TDT Raise the Bar: Project Sneak Peek																		
TDE Live Demo and Project Rollout																		
Project Deployment																		
UTDI Head of Department: Project Presentation																		
TDT Awards 2025: Copilot Award Winner																		
IP REPORT																		
IP Report preparation																		
IP Report submission																		
IP PRESENTATION																		
IP Presentation slides preparation																		
Venue booking for presentation																		
Sharing session with UTP and HC supervisor about project																		

5.5.2 Key Milestones

JULY	- Problem is identified - Objective and scope of project is set - Created a structured project timeline for progress tracking - Project idea is finalised
AUGUST	- Kickstarted the development of the Adoption Virtual Assistant (AVA) - Began development of Child Agents: Beginner, Intermediate, Matured - Presented a sneak peek of AVA at TDT Huddle: Raise the Bar
SEPTEMBER	- Submitted SIP Project title - Received feedback on develop agent and rectifying existing errors - Attended official Microsoft workshops to gain more knowledge on how to make full use of Microsoft Copilot Studio - Successfully conducted a Live Demo and Project Rollout with UTDI, effectively deploying AVA for use by UTDI personnel
OCTOBER	- Successfully presented AVA to the UTDI Head of Department, communicating its objective, features and way forward for UTDI - Received an award and recognition at the TDT Awards 2025: Copilot Award
NOVEMBER	- Successfully submitted SIP Report - Successfully shared and presented SIP Project to HC and UTP supervisor

Figure 7 : AVA Development Gantt Chart and Project Key Milestones

6.0 Result and Discussion

6.1 Adoption Virtual Assistant (AVA) Design and Functionalities

6.1.2 Adoption Virtual Assistant (AVA) System Design

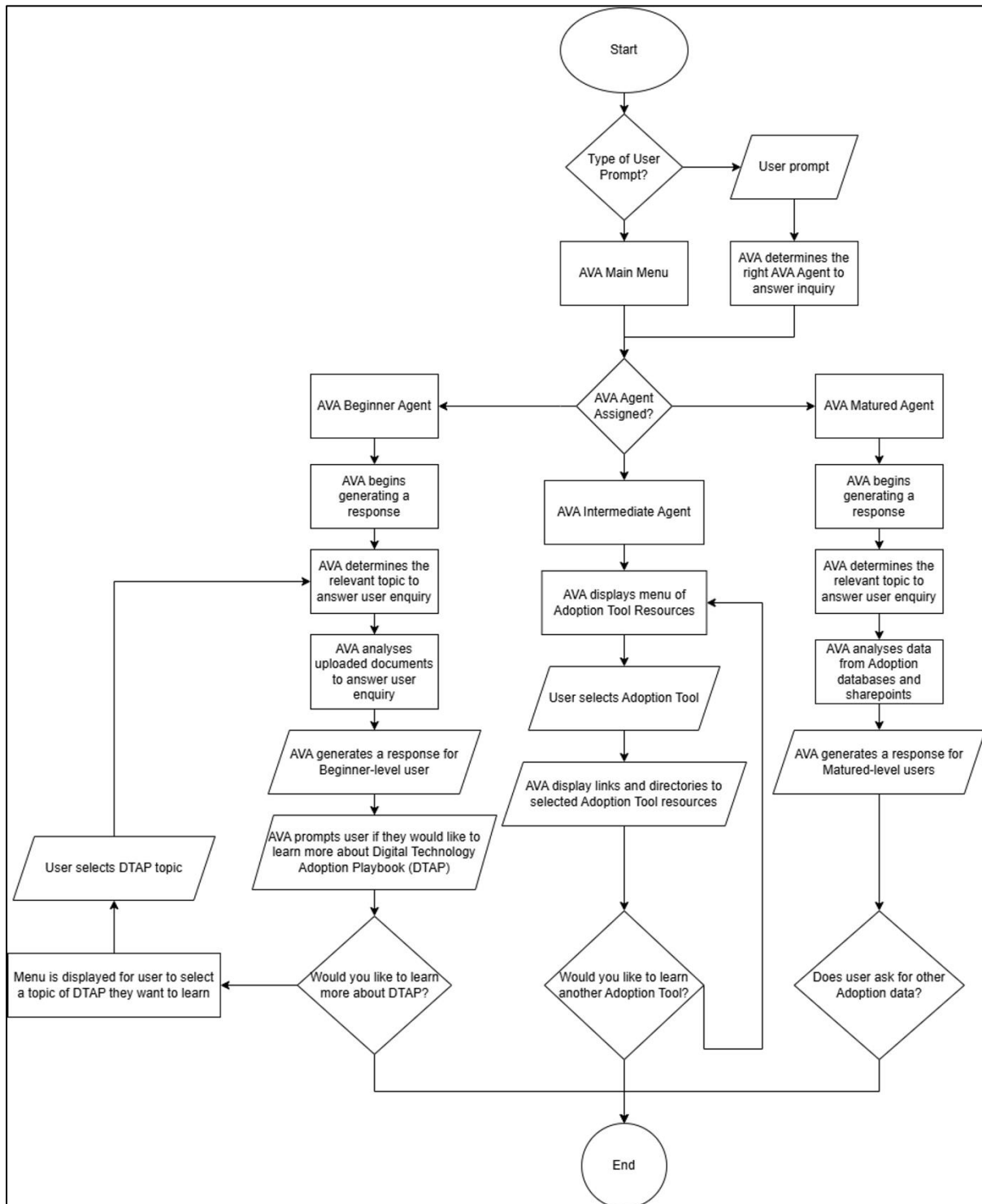


Figure 8: AVA Flowchart

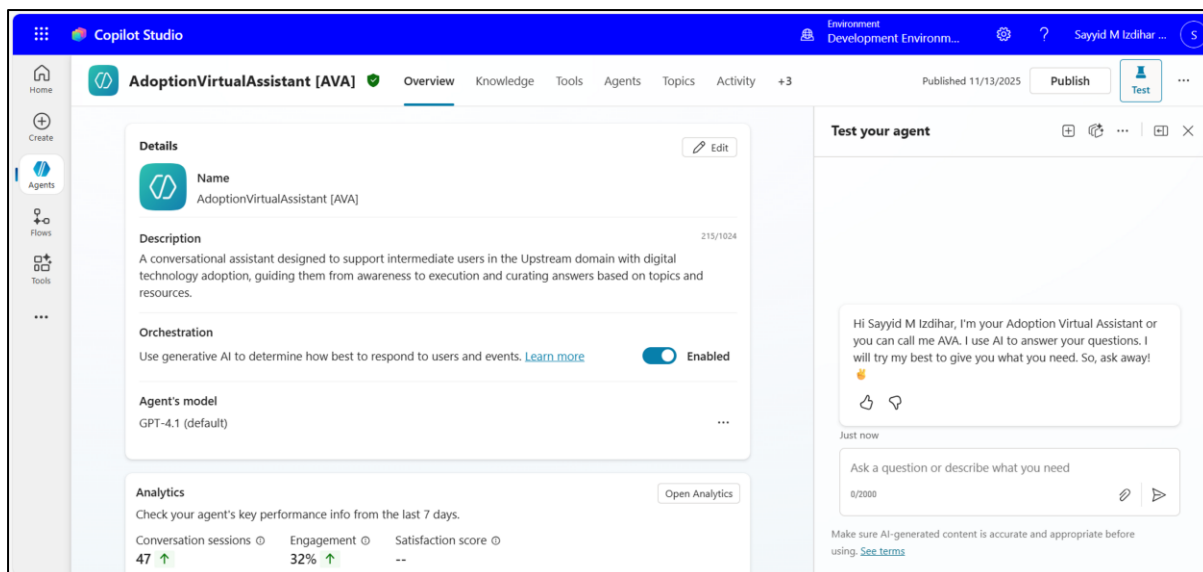
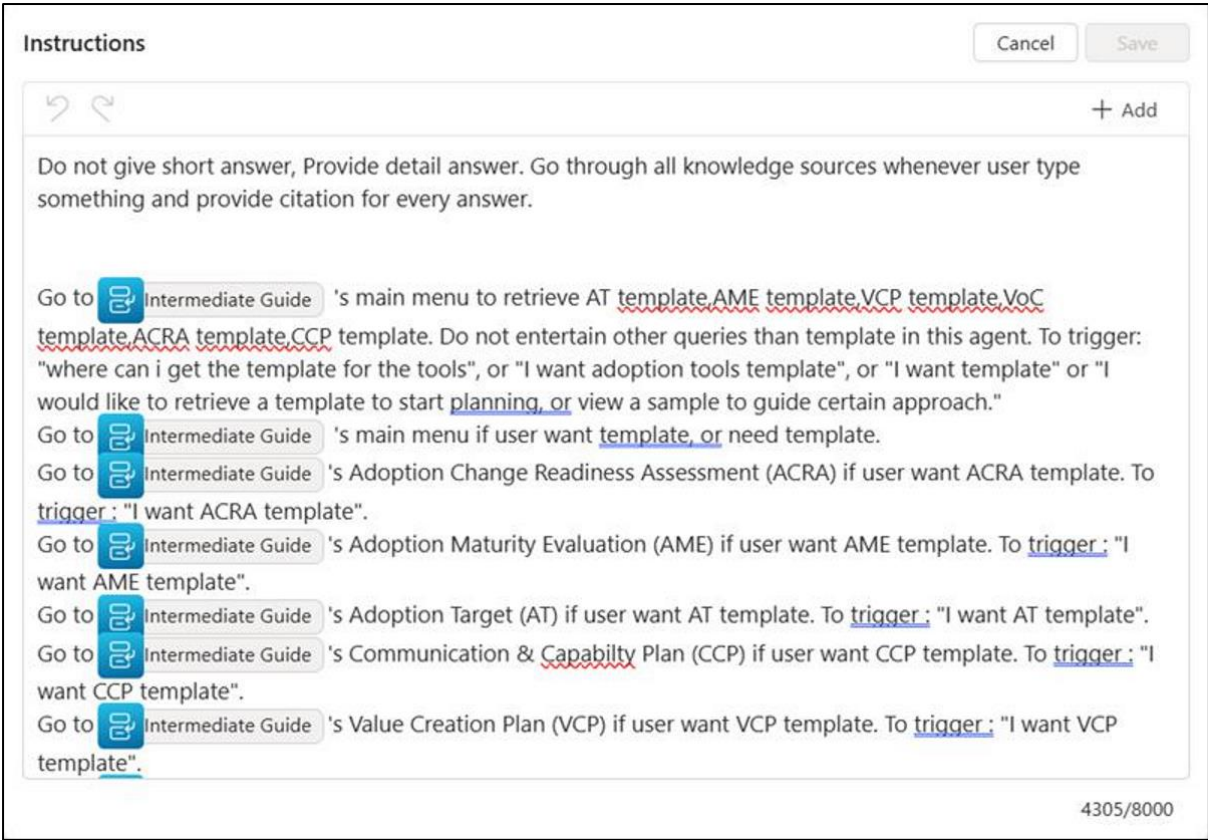


Figure 9: AVA Overview Tab on Microsoft Copilot Studio

The Overview tab in the Adoption Virtual Assistant (AVA) Developer Dashboard provides developers with a space to define the AI Agent's purpose and overall behavior. In this tab, developers can write a detailed, high-level description of what the agent is intended to accomplish, effectively guiding its decision-making and response patterns. When Orchestration is enabled, the AI Agent can utilize generative AI to determine the most suitable way to respond to user inquiries, drawing on its large language model capabilities. Developers can also select the AI Agent's model in this tab, which is currently set to GPT-4.1, offering advanced understanding, reasoning, and response generation. Essentially, the Overview tab sets the foundation for how the AI Agent interprets instructions, interacts with users, and applies its capabilities in real-world scenarios.

The Analytics tab complements this by offering insights into the agent's performance and user interactions. It acts as a feedback system, tracking metrics such as response accuracy, user engagement, response times, and common query patterns. These insights help developers evaluate the effectiveness of the AI Agent, identify areas for improvement, and make informed updates to enhance its overall functionality and user experience. By providing a continuous loop of performance data, the Analytics tab enables developers to refine the agent iteratively, ensuring it delivers accurate, contextually relevant, and user-friendly interactions over time.

Additionally, the dashboard allows developers to integrate custom configurations and settings that tailor the AI Agent to the specific needs of the organization or project. This includes defining specialized instructions, setting priority areas for responses, and linking the agent to relevant data sources within the enterprise environment. Such flexibility ensures that the agent is not only capable of general conversational tasks but also equipped to handle specialized workflows and domain-specific queries. Through this combination of descriptive guidance, performance monitoring, and customizable configuration, the AVA Developer Dashboard provides a comprehensive platform for building, managing, and continuously improving AI Agents.



The screenshot shows the 'Instructions' tab in the AVA Developer Dashboard. It features a 'Cancel' and 'Save' button at the top right. Below the title bar, there are undo and redo icons on the left and an '+ Add' button on the right. The main area contains a list of instructions for the AI agent, each starting with 'Go to [Intermediate Guide]' and followed by specific instructions and triggers. The instructions are as follows:

- Do not give short answer, Provide detail answer. Go through all knowledge sources whenever user type something and provide citation for every answer.
- Go to [Intermediate Guide] 's main menu to retrieve AT template, AME template, VCP template, VoC template, ACRA template, CCP template. Do not entertain other queries than template in this agent. To trigger: "where can i get the template for the tools", or "I want adoption tools template", or "I want template" or "I would like to retrieve a template to start planning, or view a sample to guide certain approach."
- Go to [Intermediate Guide] 's main menu if user want template, or need template.
- Go to [Intermediate Guide] 's Adoption Change Readiness Assessment (ACRA) if user want ACRA template. To trigger: "I want ACRA template".
- Go to [Intermediate Guide] 's Adoption Maturity Evaluation (AME) if user want AME template. To trigger: "I want AME template".
- Go to [Intermediate Guide] 's Adoption Target (AT) if user want AT template. To trigger: "I want AT template".
- Go to [Intermediate Guide] 's Communication & Capability Plan (CCP) if user want CCP template. To trigger: "I want CCP template".
- Go to [Intermediate Guide] 's Value Creation Plan (VCP) if user want VCP template. To trigger: "I want VCP template".

The ID 4305/8000 is displayed at the bottom right of the instructions list.

Figure 10: AVA Agent Instructions Tab

The Instructions tab in the Adoption Virtual Assistant (AVA) provides developers with a dedicated space to further refine and shape the behavior of the AI Agent. In this section, developers can give detailed guidance on how the agent should respond to different user inputs, including directing it toward particular topics or areas of focus when specific prompts are received. The tab also allows developers to configure customized responses and outputs, ensuring that the agent can address user queries in a manner that is accurate, relevant, and aligned with the goals of the organization.

A major feature of the Instructions tab is its use of natural human language to set instructions. This makes the process more accessible and easier to use, allowing developers to define complex behaviors without requiring extensive knowledge of traditional programming or coding languages. By using plain language to provide instructions, developers can quickly implement changes, test different response approaches, and adjust the agent’s behavior more efficiently.

In addition, the Instructions tab supports context-sensitive guidance. This means that the agent can adapt its responses according to the specific needs of individual users or particular situations. This ability enhances the agent’s versatility, enabling it to handle a wide variety of interactions from simple informational questions to more complex and specialized tasks. Overall, the Instructions tab is a vital tool for developers, giving them the power to shape the AI Agent’s behavior, improve its responsiveness, and ensure that it consistently provides helpful and contextually appropriate responses to users.

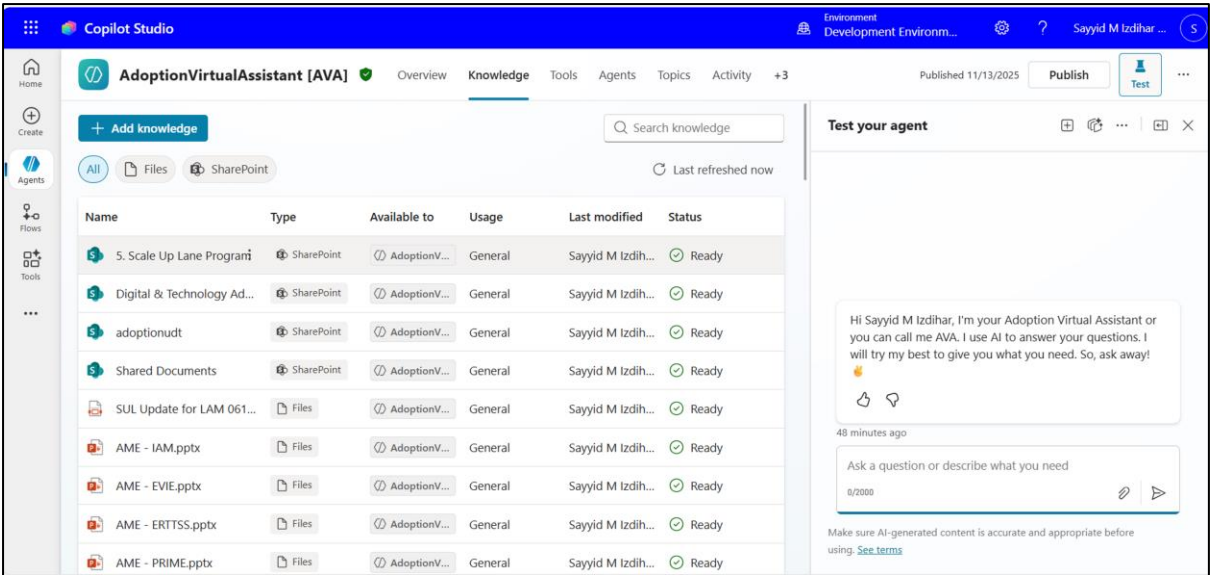


Figure 11: AVA Knowledge Tab

The Knowledge tab in the Adoption Virtual Assistant (AVA) provides developers with the ability to upload and manage documents that the AI Agent can reference when addressing user inquiries. This feature allows the agent to access a wide range of information sources, which may include files in formats such as PowerPoint, Excel, PDF, as well as content from SharePoint sites and Microsoft Lists. By incorporating these documents, the AI Agent is able to draw from authoritative and organization-specific sources, ensuring that its responses are both accurate and reliable.

When a user submits a query, the AI Agent first analyses the content of the uploaded Knowledge documents in accordance with the developer’s instructions. It processes the relevant information, identifies key points, and synthesizes the data before formulating a response. This approach guarantees that the answers provided are not generic, but are instead grounded in the official UTDI Adoption documents and internal records.

Additionally, the Knowledge tab allows developers to continuously update and expand the repository of reference materials. This flexibility ensures that the AI Agent can stay current with new information, changes in policies, or updates to procedures, maintaining the relevance and accuracy of its responses over time. By centralizing and structuring the source material, this tab enhances the AI Agent’s ability to provide informed, contextually appropriate, and highly reliable answers, making it a critical component of the AVA system for supporting both developers and end users.

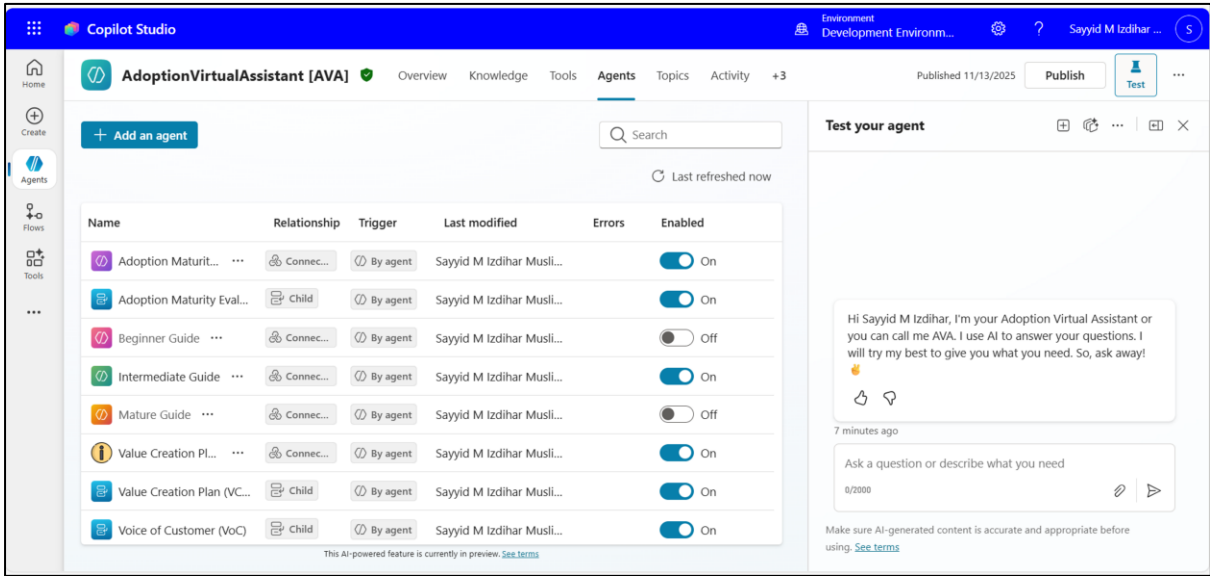


Figure 12: AVA Agents Tab

The Agents tab in the Adoption Virtual Assistant (AVA) provides a clear view of the hierarchical structure of the AI system, showing that AVA functions as the parent agent overseeing multiple specialized child agents. These child agents include the Beginner Guide, Intermediate Guide, and Matured Guide, each designed with a distinct set of instructions and a dedicated knowledge base. This separation ensures that the AI maintains accuracy and consistency in its responses while minimizing the risk of hallucinations or confusion between different topics.

When a user submits a query, the parent agent first analyses the nature, type, and context of the question. Based on this initial assessment, the parent agent determines which child agent is best equipped to handle the inquiry. The selected child agent then processes the query using its specific scope of knowledge and pre-defined instructions, generating a response that is tailored, precise, and aligned with the intended purpose of that agent.

This modular approach allows the Adoption Virtual Assistant to handle a wide variety of questions efficiently and reliably, providing users with targeted and contextually appropriate information. It also allows developers to manage and update each child agent independently, ensuring that changes in one area do not inadvertently affect other areas. Overall, the Agents tab illustrates how the AVA system combines structured organization, specialized expertise, and intelligent routing to deliver accurate and well-informed responses, making the interaction more reliable and user-focused.

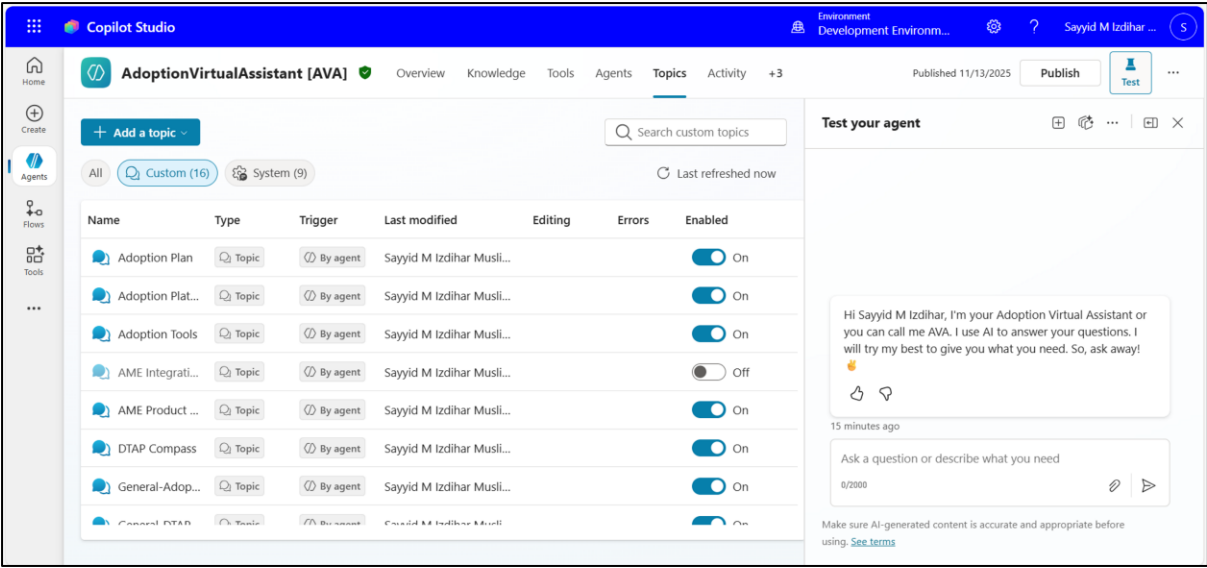


Figure 13: AVA Topics Tab

The Topics tab in the Adoption Virtual Assistant (AVA) provides developers with a powerful tool to further refine and tailor the behavior of both parent and child agents. Within this tab, developers can define the specific subjects or areas of focus that each agent is allowed to address, effectively guiding the scope of its interactions with users. This targeted configuration ensures that the AI Agent responds appropriately and remains relevant to the intended domain of knowledge, improving the accuracy and usefulness of its responses.

The topics defined in this tab are closely linked to the knowledge documents uploaded in the Knowledge tab, allowing the agent to reference verified and authoritative sources when generating answers. By connecting topics to specific reference materials, developers can prevent the AI Agent from providing misleading or irrelevant information, maintaining a high standard of reliability in its interactions.

In addition, configuring topics allows developers to exercise granular control over how the AI Agent interprets user inquiries and prioritizes information. This makes it possible to design an agent that is context-aware and capable of handling queries in a structured and organized manner. The Topics tab also supports ongoing updates and adjustments, enabling developers to expand or refine the agent’s areas of expertise over time as new knowledge documents are added or organizational needs evolve. Overall, the Topics tab plays a critical role in shaping the intelligence and functionality of the AI Agent, ensuring that it consistently delivers accurate, contextually appropriate, and highly relevant responses to user queries.

6.1.3 Adoption Virtual Assistant Functionalities and Features

The Adoption Virtual Assistant (AVA) was specifically designed with advanced functionalities to bridge the knowledge gap in Digital Technology Adoption within the organization. It categorizes users into three distinct levels; Beginner, Intermediate, and Matured to provide guidance and responses that are tailored to each user's level of proficiency. By adjusting the complexity, depth, and specificity of its responses according to the user's knowledge and experience, AVA ensures that all users receive support that is relevant and actionable, whether they are just starting their digital adoption journey or are already experienced practitioners.

To manage specialized queries effectively, the system incorporates three dedicated Child Agents, each focused on handling inquiries within their defined scope. These agents are supported by a structured framework of 33 conversational topics, allowing the AI to navigate user questions efficiently, and are connected to 68 verified documents sourced from the UTDI Adoption SharePoint. This integration ensures that the responses provided are accurate, reliable, and grounded in official organizational knowledge, promoting data-driven decision making and minimizing the risk of misinformation.

AVA guides users through the Digital Technology Adoption process using the Digital Adoption Playbook (DTAP), providing foundational support for those seeking to initiate or accelerate their adoption journey. For users who require assistance in applying adoption tools, AVA offers detailed guidance on utilizing templates, applications, and dashboards effectively. Beyond serving as a source of instruction, AVA also acts as a personal assistant for more experienced users, delivering real-time insights and performance data. It can generate comprehensive reports, provide actionable feedback, and recommend strategies to optimize the use of digital technology within PETRONAS Upstream operations.

By combining user segmentation, structured knowledge resources, and advanced AI capabilities, AVA not only facilitates learning and adoption but also supports informed decision-making, enhances operational efficiency, and ensures a consistent, high-quality digital technology adoption experience across the organization. This multifaceted approach positions AVA as a strategic tool that empowers users at all levels to navigate, implement, and maximize the value of digital technologies in a systematic and effective manner.

AVA Beginner Agent Functionalities and Features

Beginner Guide

OverviewKnowledgeToolsAgentsTopicsActivityEvaluationAnalyticsChannels

Published 8/27/2025PublishSettingsTest

+ Add a topic

AllCustom (15)System (9)

Search custom topics

Last refreshed now

Name	Type	Trigger	Last modified	Editing	Errors	Enabled
General-Adoption (Beginner)	Topic	By agent	Sayyid M Izzihar Muslimin (Dev/Upstream)...			On
General-DTAP (Beginner)	Topic	By agent	Sayyid M Izzihar Muslimin (Dev/Upstream)...			On
Getting Started With Adoption (Beginner)	Topic	By agent	Sayyid M Izzihar Muslimin (Dev/Upstream)...			Off
Goodbye	Topic	By agent	Sayyid M Izzihar Muslimin (Dev/Upstream)...			On
Greeting	Topic	By agent	Sayyid M Izzihar Muslimin (Dev/Upstream)...			On
Intro1	Topic	By agent	Sayyid M Izzihar Muslimin (Dev/Upstream)...			Off
Main Menu	Topic	By agent	Sayyid M Izzihar Muslimin (Dev/Upstream)...			On
Start Over	Topic	By agent	Sayyid M Izzihar Muslimin (Dev/Upstream)...			On
T1.1 General Understanding of DTAP	Topic	By agent	Sayyid M Izzihar Muslimin (Dev/Upstream)...			On
T1.2 DTAP Compass	Topic	By agent	Sayyid M Izzihar Muslimin (Dev/Upstream)...			On
T1.3 Adoption Plan	Topic	By agent	Sayyid M Izzihar Muslimin (Dev/Upstream)...			On

Figure 14: AVA Beginner Topics Overview

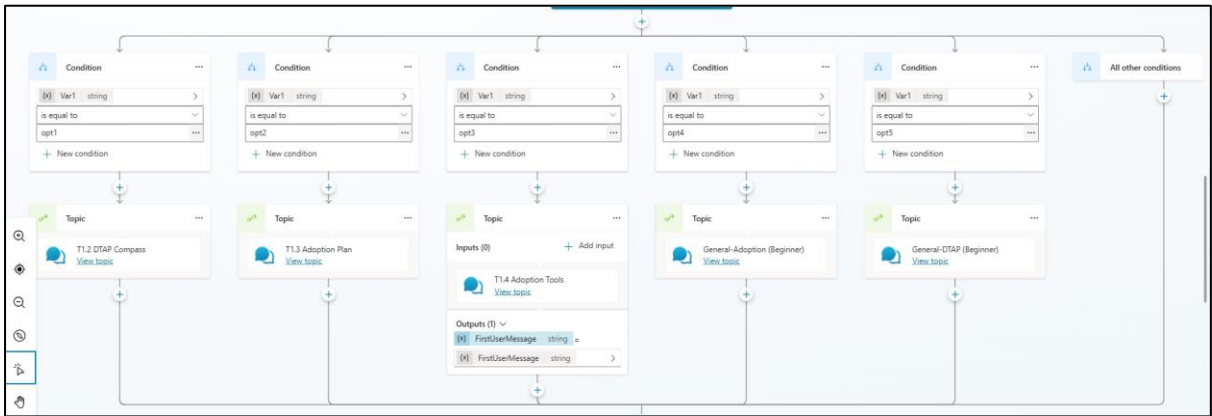


Figure 15: AVA Beginner Topics Tree

Adaptive Card

What would you like to explore today?
Please choose a topic below so I can guide you to the right section of DTAP 2.0:

☐ DTAP Compass

☐ Adoption Plan

☐ Adoption Tools

☐ What's Adoption

☐ What's DTAP?

Submit

Outputs (3) ▾

{x} actionSubmitId	string	=	actionSubmitId
{x} id	string	=	id
{x} Var1	string	=	Var1

Figure 16: AVA Beginner Topics Node (Beginner Menu)

The Beginner Agent within the Adoption Virtual Assistant (AVA) system was specifically developed to provide foundational support and guidance to users who are new to Digital Technology Adoption, particularly within the context of PETRONAS Upstream operations and the Upstream Technology Digital Innovation (UTDI) department. This agent acts as the primary entry point for personnel with minimal or no prior knowledge of digital technology adoption concepts and processes. Its primary objective is to introduce users to essential principles and ideas, ensuring that they develop a solid conceptual understanding before progressing to more complex or advanced topics. By providing this structured introductory support, the Beginner Agent helps users gain confidence and clarity as they begin their digital adoption journey.

To enhance understanding and engagement, the Beginner Agent is designed to deliver responses that include visual aids, such as illustrative diagrams and interactive elements, alongside dynamic generative answers. These responses are created by carefully analysing and synthesizing information from a variety of uploaded knowledge sources, including verified PDFs, Excel spreadsheets, and PowerPoint presentations from the UTDI SharePoint repository. This ensures that the guidance provided is accurate, data-driven, and aligned with organizational standards.

The interaction with the Beginner Agent is further structured through a predetermined menu system that guides users through available topics and offers clear options for exploration. For example, when a user asks a basic question such as "What is Digital Technology Adoption?", the agent identifies the inquiry as suitable for the Beginner level. It then navigates through its predefined topic nodes, which are organized like decision trees, to categorize the query and route it appropriately. Once the relevant topic is identified, the agent generates a tailored response based on its knowledge base, presenting information in a clear and structured manner to facilitate understanding.

Furthermore, the Beginner Agent not only provides explanations but also encourages interactive learning by offering step-by-step guidance and examples that relate directly to the user's operational context. This ensures that users are not only introduced to theoretical concepts but also understand their practical applications within PETRONAS Upstream. Through this combination of visual aids, structured menus, and contextually relevant knowledge, the Beginner Agent serves as a comprehensive learning companion, empowering new users to build a strong foundation in Digital Technology Adoption and preparing them for more advanced engagement with the Intermediate and Matured Agents within the AVA system.

AVA Intermediate Agent Functionalities and Features

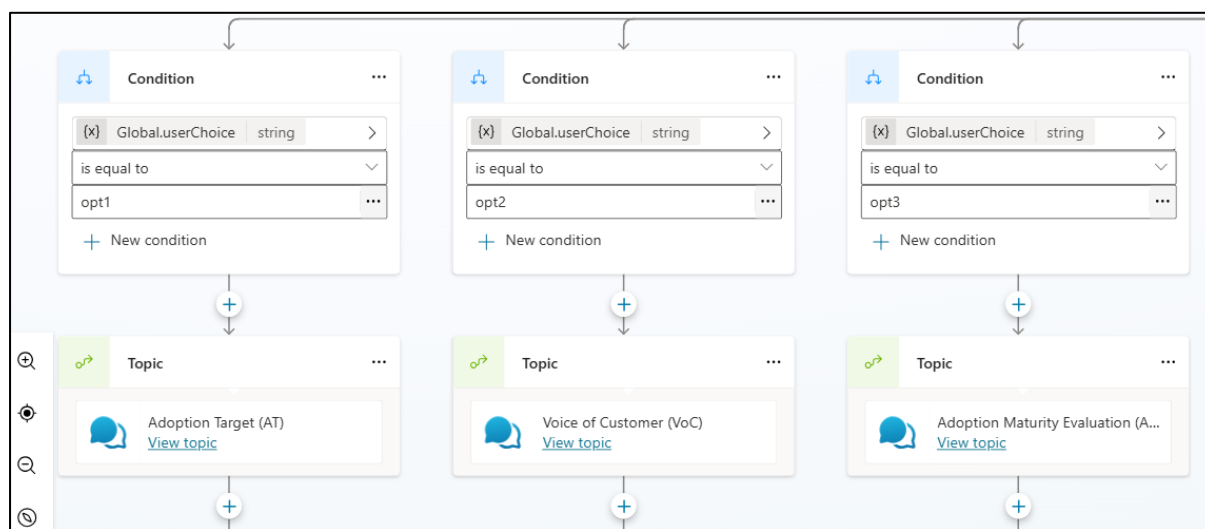


Figure 17: AVA Intermediate Topics Tree

Adaptive Card

Which template should I pull up for you?

☐

Adoption Target (AT)

☐

Voice of Customer (VoC)

☐

Adoption Maturity Evaluation (AME)

☐

Value Creation Plan (VCP)

☐

Adoption Change Readiness Assessment (ACRA)

☐

Capability & Change Management Plan (CCP)

☐

I do not know which tool to choose... 🤔

Submit

Outputs (3) ▾

(x)

actionSubmitId

string

=

(x)

actionSubmitId

string

>

(x)

id

string

=

Figure 18: AVA Intermediate Topics Node (Intermediate Menu)

Condition

(x)

userChoice

string

>

is equal to ▾

opt1 ▾

+ New condition

+

Message

Text ▾

+ Add ▾

B

I

≡

≡

(x)

fx

Here is your AME sample!

• [AME - PROTEAN.pptx]

(https://petronas.sharepoint.com/:p/r/sites/adoptionudt/Shared%20Documents/2.%20Alignment/4.%20VoC,%20AME%20%26%20Value/2024/3.%20AME/PROTEAN/AME%20-%20PROTEAN.pptx?d=w3c952d6ad0ba456097e4f9a0360addad&csf=1&web=1&e=PuWyEG)

Figure 19: AVA Intermediate Topics Node

The Intermediate Agent within the Adoption Virtual Assistant (AVA) system is designed to support users who have already acquired a basic understanding of Digital Technology Adoption and are seeking to deepen and refine their knowledge. Unlike the Beginner Agent, the Intermediate Agent does not focus on providing elementary explanations or introductory concepts. Instead, it emphasizes practical, actionable guidance and focuses on helping users apply their existing knowledge to real-world tasks and scenarios. This approach ensures that users at this level can progress efficiently, without revisiting content they have already mastered.

The Intermediate Agent plays a critical role in assisting users who are in the process of initiating or enhancing their Digital Technology Adoption journey. It offers detailed support in navigating and effectively utilizing specialized Adoption Tools, ensuring that users can extract maximum value from each tool. These tools include the Adoption Target (AT), which helps define and track adoption goals; the Voice of Customer (VoC), which gathers and analyses user feedback; the Adoption Maturity Evaluation (AME), which assesses the current level of adoption within processes; the Value Creation Plan (VCP), which helps identify potential benefits and opportunities; the Adoption Change Readiness Assessment (ACRA), which evaluates preparedness for adopting new technologies; and the Capability and Change Management Plan (CCP), which provides a structured approach to implementing and sustaining digital initiatives.

By focusing on these practical applications, the Intermediate Agent ensures that users are not only gaining theoretical insights but are also equipped with the guidance needed to apply adoption principles effectively within their organizational context. The agent provides step-by-step instructions, best practices, and context-aware recommendations, enabling users to make informed decisions and take meaningful actions that advance their adoption maturity.

Additionally, the Intermediate Agent is capable of synthesizing information from the AVA knowledge base, including verified documents, templates, and tools, to generate responses that are accurate, relevant, and actionable. Its design encourages proactive learning, problem-solving, and the efficient use of digital technology adoption resources, bridging the gap between foundational understanding and advanced mastery. Overall, this agent empowers users to move confidently from basic knowledge to more sophisticated application, positioning them to maximize the effectiveness of digital adoption efforts within PETRONAS Upstream operations.

AVA Matured Agent Functionalities and Features

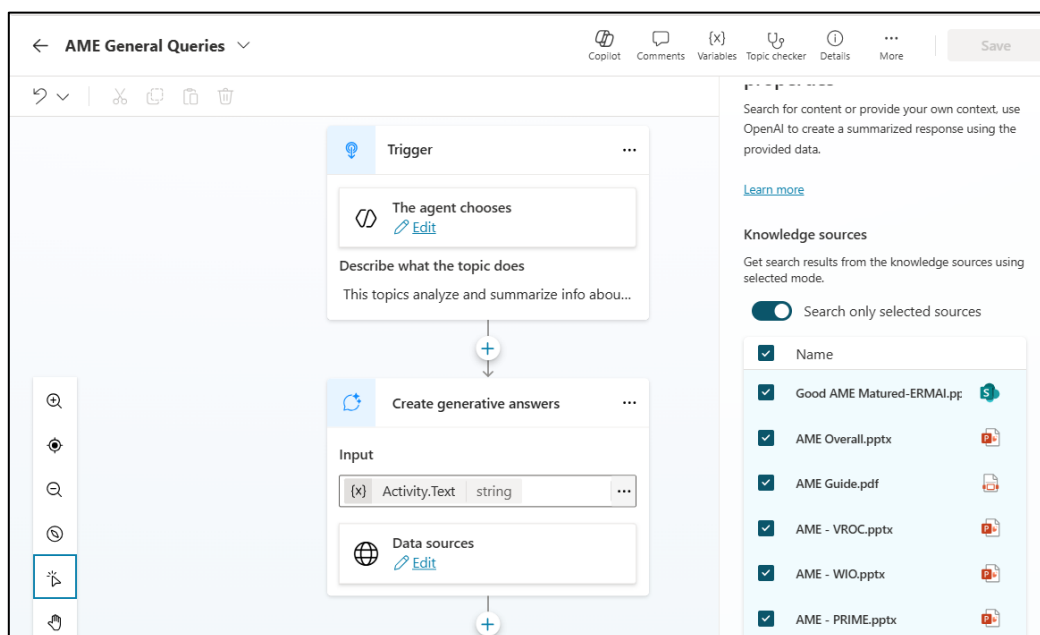


Figure 20: AVA Matured Agent Topics Node with Knowledge Source selected

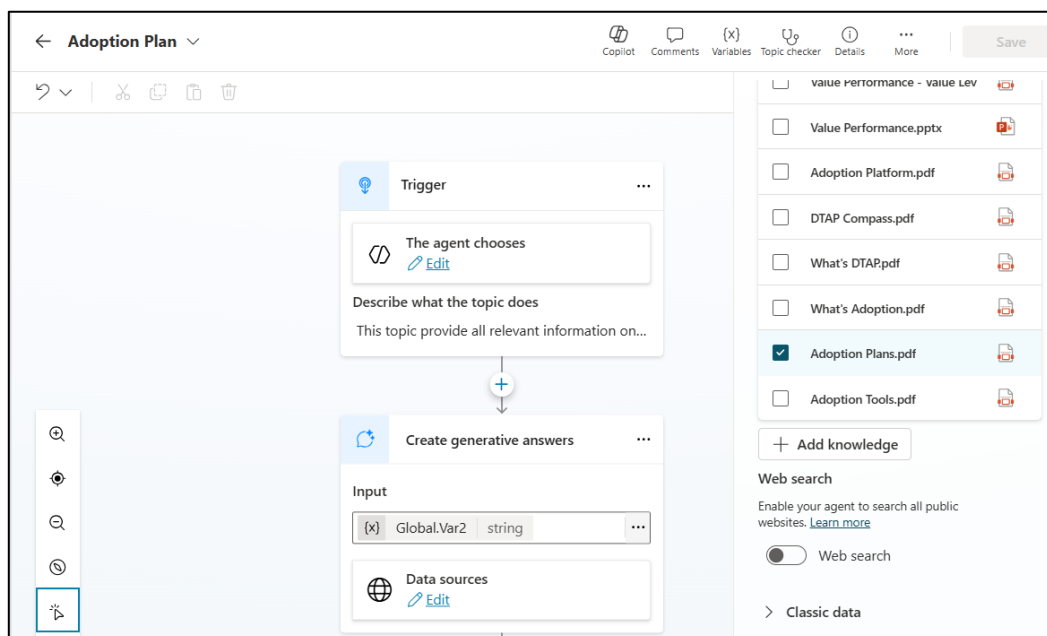


Figure 21: AVA Matured Agent Topics Node with Knowledge Source (Adoption Plan)

The Matured Agent within the Adoption Virtual Assistant (AVA) system is specifically designed for users who possess advanced knowledge and experience in Digital Technology Adoption processes and are seeking a sophisticated tool to optimize the efficiency of their adoption journey. Unlike tools aimed at beginners or intermediates, this agent is tailored for professionals who already have a strong understanding of both foundational and intermediate concepts and wish to further refine their strategies, streamline repetitive tasks, and achieve superior outcomes with minimal operational friction. By focusing on high-level productivity,

the Matured Agent allows users to dedicate more time to strategic decision-making and complex problem-solving rather than routine navigation or basic adoption steps.

One of the key functionalities of the Matured Agent is its ability to monitor and assess a product's performance in terms of user adoption. It achieves this by analyzing data from uploaded documents, SharePoint repositories, and PowerApps databases to extract critical metrics such as the current Customer Satisfaction Score (CSAT) and the Adoption Maturity Level. This analytical capability enables the agent to not only measure the effectiveness of a product in real-world deployment but also to provide actionable insights for continuous improvement. Based on these analyses, the agent generates recommendations for forward-looking strategies, potential product enhancements, and risk mitigation measures, ultimately supporting the user in achieving sustained success in their digital initiatives.

The Matured Agent places particular emphasis on guiding users through the most essential Adoption Tools, including the Adoption Maturity Evaluation (AME), the Value Creation Plan (VCP), and the Voice of Customer (VoC). These tools are crucial in providing a comprehensive understanding of adoption progress, value generation, and end-user feedback, thereby empowering users to make informed decisions and implement targeted strategies.

To ensure highly personalized and precise support, the Matured Agent employs generative responses that are informed by a deep analysis of uploaded documents stored within its knowledge repository. It also interprets the embedded instructions within its pre-configured topic nodes, which contain expert-level prompts and logic flows. This capability enables the agent to synthesize complex information into highly detailed, context-specific responses that address the user's unique questions with accuracy and relevance. By combining advanced data analysis with expert-guided knowledge processing, the Matured Agent becomes an indispensable resource for experienced users seeking to elevate their digital technology adoption efforts, optimize outcomes, and maintain long-term success in an increasingly digitalized environment.

6.2 Adoption Virtual Assistant (AVA) Output Evaluation

AVA Main Menu

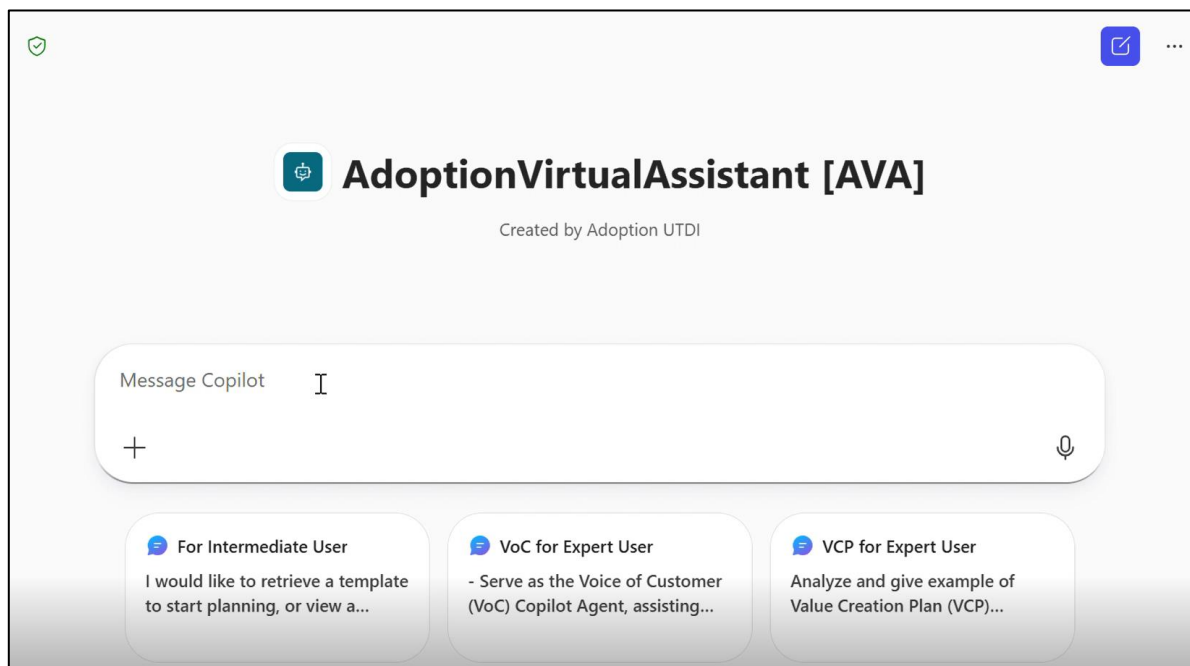


Figure 22: AVA Main Menu on Deployed Platform

The Main Menu of the Adoption Virtual Assistant (AVA) serves as the central interface through which users interact with the system. At the forefront of this interface is a chat box, enabling users to input their questions, requests, or any specific queries they may have regarding digital technology adoption. This chat functionality allows AVA to provide immediate, context-aware responses, facilitating a smooth and interactive user experience.

In addition to the chat box, the Main Menu offers a set of suggestion buttons designed to guide users toward relevant resources and functionalities. These buttons include options such as “For Intermediate User,” which provides tailored support for users who already have some foundational knowledge in digital adoption, as well as “VOC for Expert User” and “VCP for Expert User,” which are geared toward advanced users seeking in-depth guidance on the Voice of Customer (VoC) and Value Creation Plan (VCP) tools, respectively. These suggestion buttons serve as a convenient navigation aid, helping users quickly access the features and guidance most relevant to their level of expertise.

For ease of use, the Beginner Agent is pre-integrated within AVA and is available by default. This ensures that users who are new to digital technology adoption can immediately receive guidance and support without the need to manually select or activate the agent. By providing this seamless access, AVA caters to a broad spectrum of users, from novices to experts, offering a tailored and efficient approach to learning and implementing digital adoption strategies.

Testing AVA Beginner Agent

What are the 3 scopes of Adoption based on DTAP

AdoptionVirtualAssistant [AVA]

Based on the Digital Technology Adoption Playbook (DTAP), the three main scopes of Adoption are:

1. **Capability:** This scope focuses on developing and uplifting users' ability to use digital products as intended. It includes capability development plans, upskilling, training, and ensuring a pool of competent or expert users. Key actions involve evaluating capability bench-strength, sharing best practices, and tracking completion of training or mentoring sessions.

Figure 23: AVA Beginner Agent Output

2. **User Experience:** This scope aims to ensure users have a positive and effective experience with the product. It involves strategies for communication, change management, and continuous evaluation of product effectiveness through metrics like Customer Satisfaction Score (CSat), utilization rate, and user feedback. It also ensures product health and maturity, keeping the product relevant to users.
3. **Value:** This scope ensures that the adoption of digital products delivers measurable value aligned with business objectives. It involves the development and implementation of Value Creation Plans (VCP), value reporting, and analysis of value performance. The focus is on accurate and timely value realization reporting and supporting the removal of deadlocks in value reporting.

Figure 24: AVA Beginner Agent Output

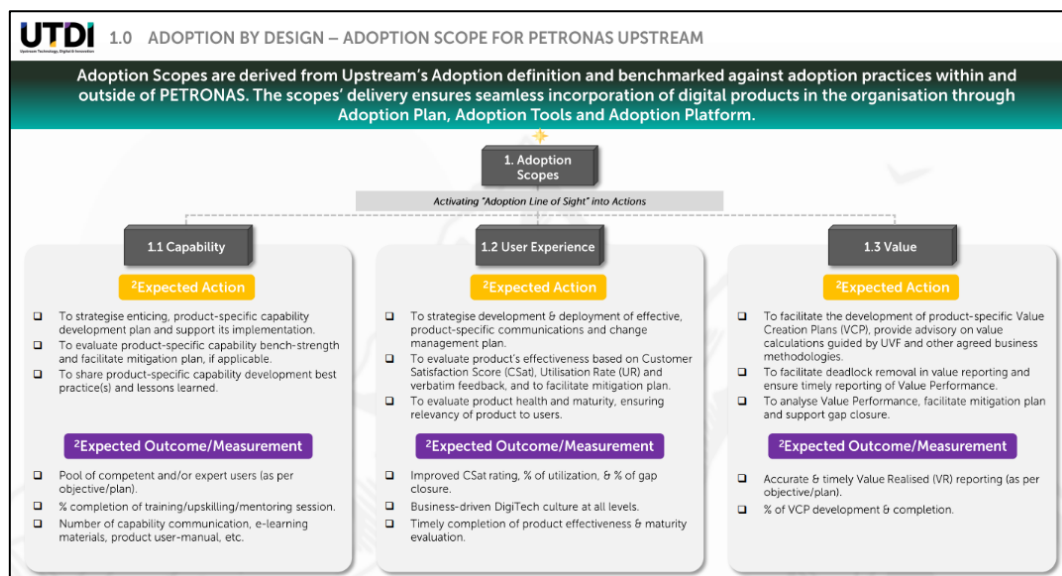
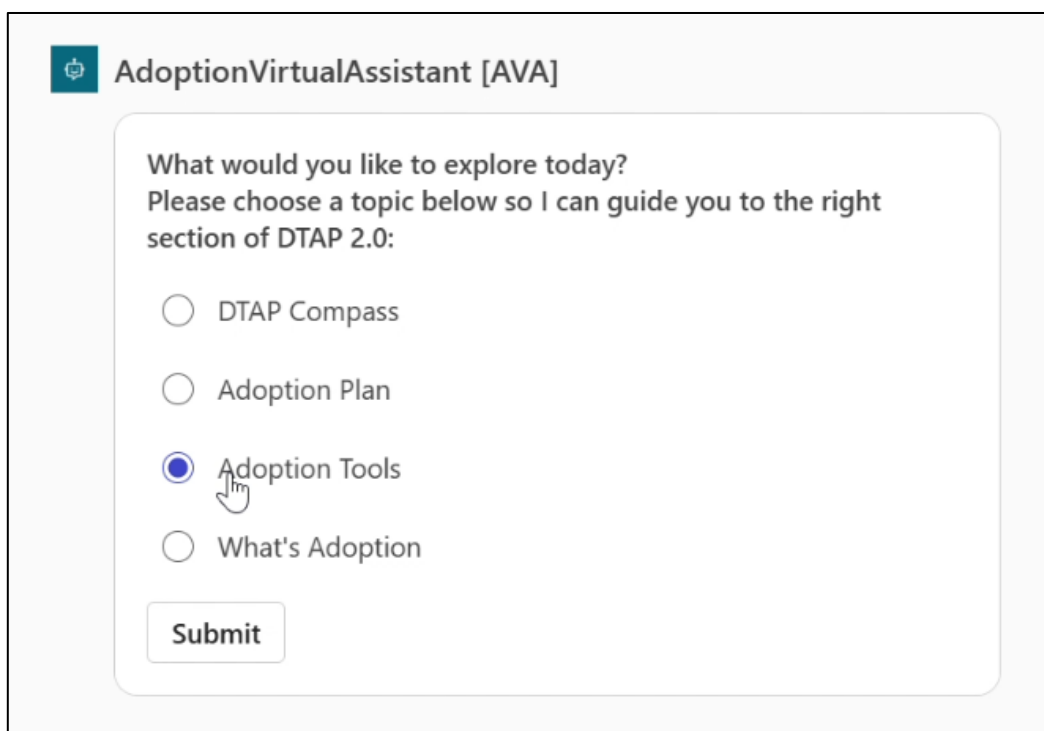


Figure 25: DTAP Document : Adoption Scopes

During the testing phase of the Adoption Virtual Assistant (AVA) Beginner Agent, a specific scenario was evaluated to assess the agent's accuracy and responsiveness. In this scenario, the user posed the question, "What are the three scopes of Adoption based on DTAP?" In response, AVA utilized its document analysis capabilities to thoroughly examine the relevant DTAP materials stored within its knowledge repository. Through this process, the agent was able to identify and synthesize the information accurately, ultimately generating a response that correctly outlined the three defined scopes of Adoption.

To verify the correctness of AVA's response, a direct comparison was conducted between the answer provided by the agent and the information presented in the Digital Adoption Playbook (DTAP). This comparison confirmed that the Beginner Agent not only retrieved the correct content but also presented it in a coherent and contextually appropriate manner. The test demonstrated the agent's ability to interpret structured documents effectively, extract key insights, and deliver precise responses to user queries. By successfully identifying the three scopes of Adoption, the AVA Beginner Agent showcased its capability to support users in understanding foundational concepts of digital technology adoption, reinforcing its role as an accessible and reliable learning tool for individuals at the beginner level.



The screenshot displays the 'AdoptionVirtualAssistant [AVA]' interface. At the top left is a blue square icon with a white robot head. To its right is the title 'AdoptionVirtualAssistant [AVA]'. Below this, a light gray rounded rectangle contains the text: 'What would you like to explore today? Please choose a topic below so I can guide you to the right section of DTAP 2.0:'. Underneath this text are four radio button options: 'DTAP Compass', 'Adoption Plan', 'Adoption Tools', and 'What's Adoption'. The 'Adoption Tools' option is selected, indicated by a blue dot and a mouse cursor icon pointing at it. At the bottom of the rounded rectangle is a 'Submit' button.

Figure 26: AVA Beginner Agent Menu Selection

Adoption Tools in the DTAP Framework

The DTAP (Design, Test, Adopt, and Platform) framework includes a comprehensive set of Adoption Tools designed to facilitate successful product deployment and management. These tools support product owners and managers in strategizing and executing adoption activities, engaging stakeholders, creating value, and managing change. Below are the key Adoption Tools outlined in the DTAP Compass:

- **Adoption Target (AT):** Helps define and set clear adoption goals for the product or project.
- **Voice of Customer (VoC):** Captures and integrates customer feedback to ensure the product meets user needs and expectations.
- **Adoption Maturity Execution (AME):** Assesses and tracks the maturity level of adoption activities, ensuring continuous improvement.
- **Value Creation Plan (VCP):** Outlines strategies for delivering and maximizing value through the adoption process.
- **Adoption Readiness Assessment (ACRA):** Evaluates the preparedness of stakeholders and systems for

Figure 27: AVA Beginner Agent : Adoption Tool Output

Following the delivery of its initial response, AVA presented the user with a supplementary menu designed to offer further learning opportunities related to DTAP 2.0. This menu included several options, such as DTAP Compass, Adoption Plan, Adoption Tool, and What’s Adoption, allowing the user to explore specific aspects of digital adoption in greater depth. When the user chose the Adoption Tool option, AVA accurately identified the selection and proceeded to generate a comprehensive list of Adoption Tools. Each tool was accompanied by a concise explanation, providing the user with clear and informative guidance on the purpose and application of each tool within the adoption process.

Sources

Citations

DTAP 2.0 - Adoption Tools.pptx
petronas.sharepoint.com

Digital & Technology Adoption Playbook (DTAP).p...
petronas.sharepoint.com

DTAP Compass.pdf

What's Adoption.pdf

Figure 28: AVA Beginner Agent : Data Source Window

Users are also given the option to access the source of the information, enabling them to confirm the origin of the details that AVA used to formulate its response.

Testing AVA Intermediate Agent

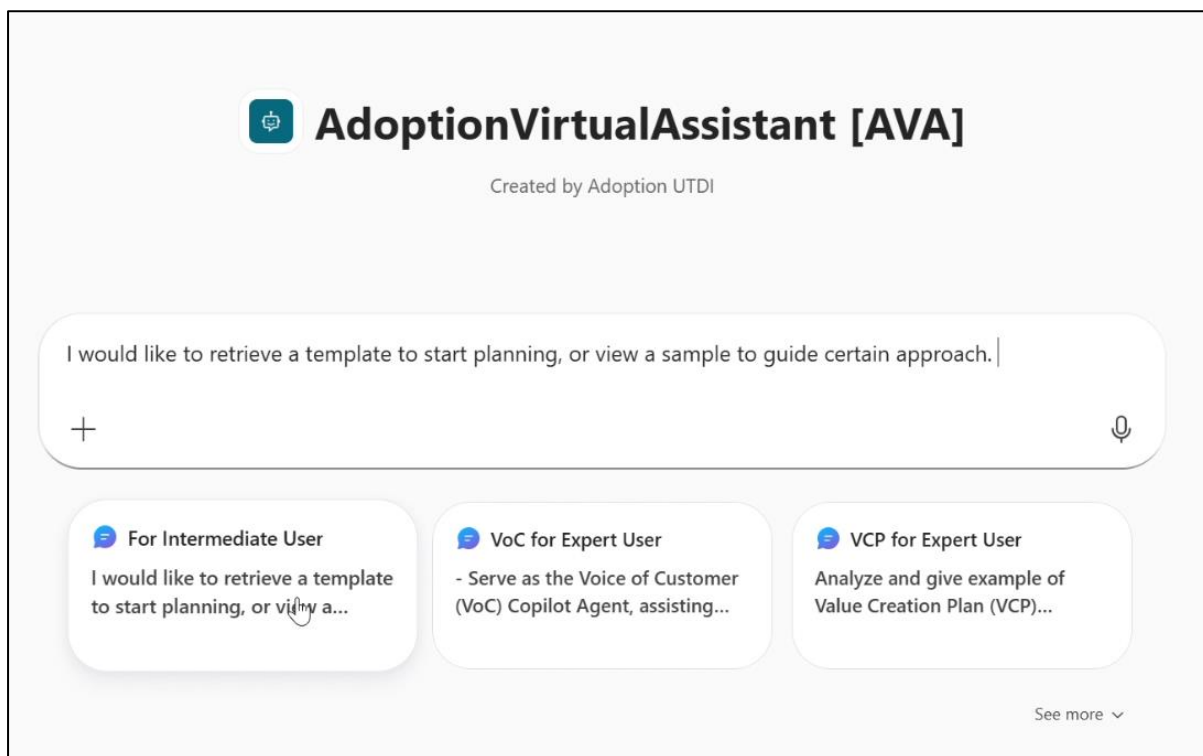


Figure 29: AVA Intermediate Agent : User selecting Intermediate Agent

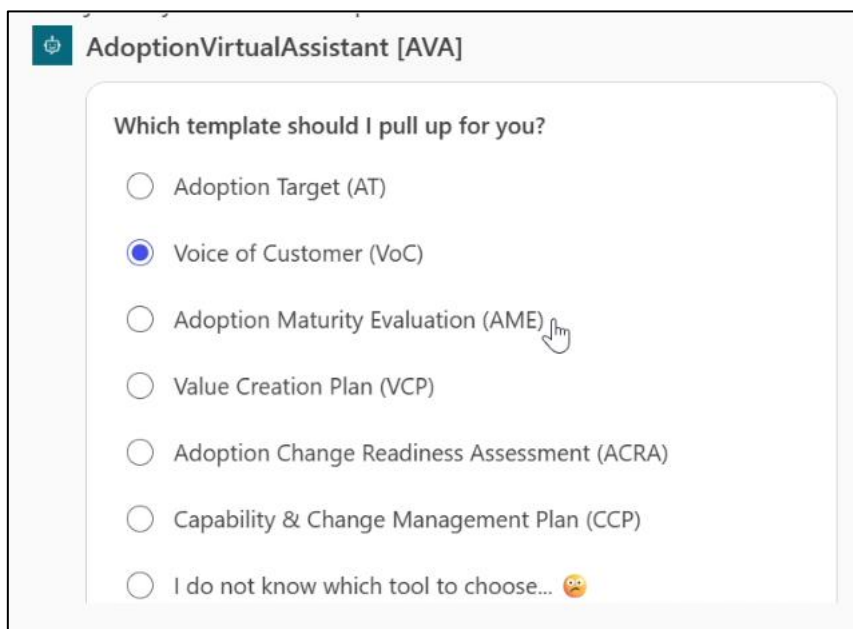
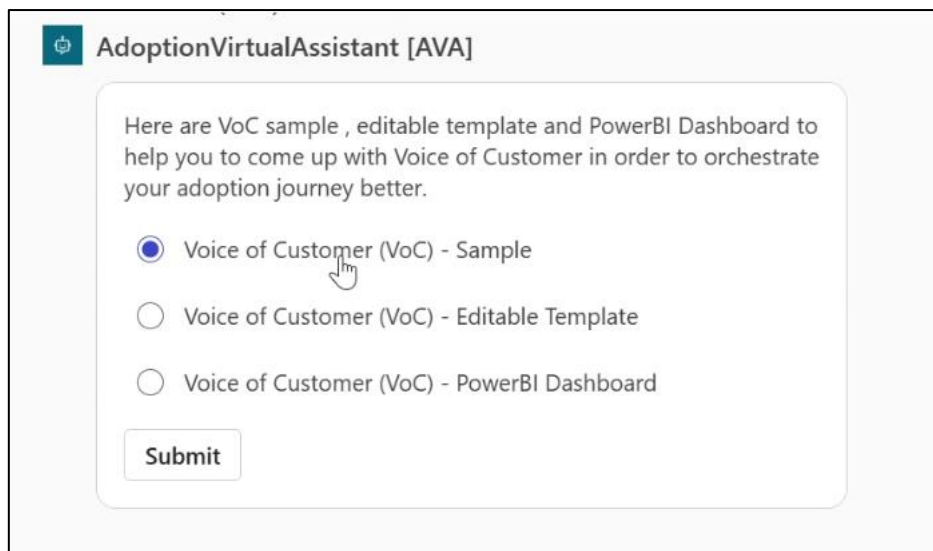


Figure 30: AVA Intermediate Agent Menu



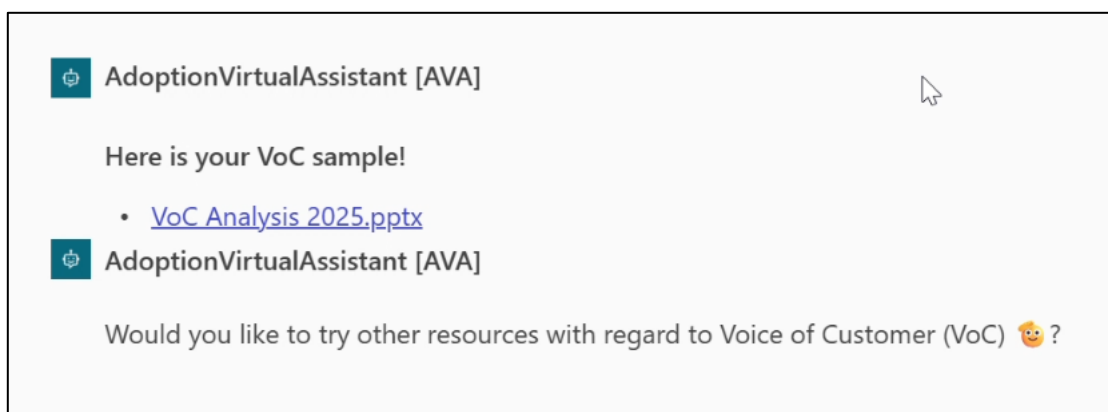
AdoptionVirtualAssistant [AVA]

Here are VoC sample , editable template and PowerBI Dashboard to help you to come up with Voice of Customer in order to orchestrate your adoption journey better.

☒ Voice of Customer (VoC) - Sample
☐ Voice of Customer (VoC) - Editable Template
☐ Voice of Customer (VoC) - PowerBI Dashboard

Submit

Figure 31: AVA Intermediate Agent VoC Sample Menu



AdoptionVirtualAssistant [AVA]

Here is your VoC sample!

- [VoC Analysis 2025.pptx](#)

AdoptionVirtualAssistant [AVA]

Would you like to try other resources with regard to Voice of Customer (VoC) 😊?

Figure 32: AVA Intermediate Agent Output

During the testing process of the AVA Intermediate Agent, the user opted to select the “For Intermediate User” option. AVA successfully recognised this selection and automatically transitioned to its Intermediate Agent mode, adjusting its responses and guidance to suit the user’s higher level of expertise. Once the Intermediate Agent was activated, the user was presented with a structured menu containing various templates designed to facilitate the effective use of the Voice of Customer (VoC) tool. These templates were curated to assist users in capturing, analysing, and leveraging customer feedback in a systematic and insightful manner.

When the user chose the Voice of Customer (VoC) sample from the menu, AVA accurately identified the selection and promptly generated a link that allowed the user to access the VoC Analysis 2025. This document provided a collection of well-organised and meaningful VoC samples, offering concrete examples that could guide the user in conducting their own customer feedback analysis. The test demonstrated the Intermediate Agent’s ability to recognise user intent, provide targeted resources, and deliver practical outputs that are directly applicable to

digital adoption practices, thereby supporting more advanced users in enhancing their adoption strategies with real-world, data-driven insights.

Testing AVA Matured Agent

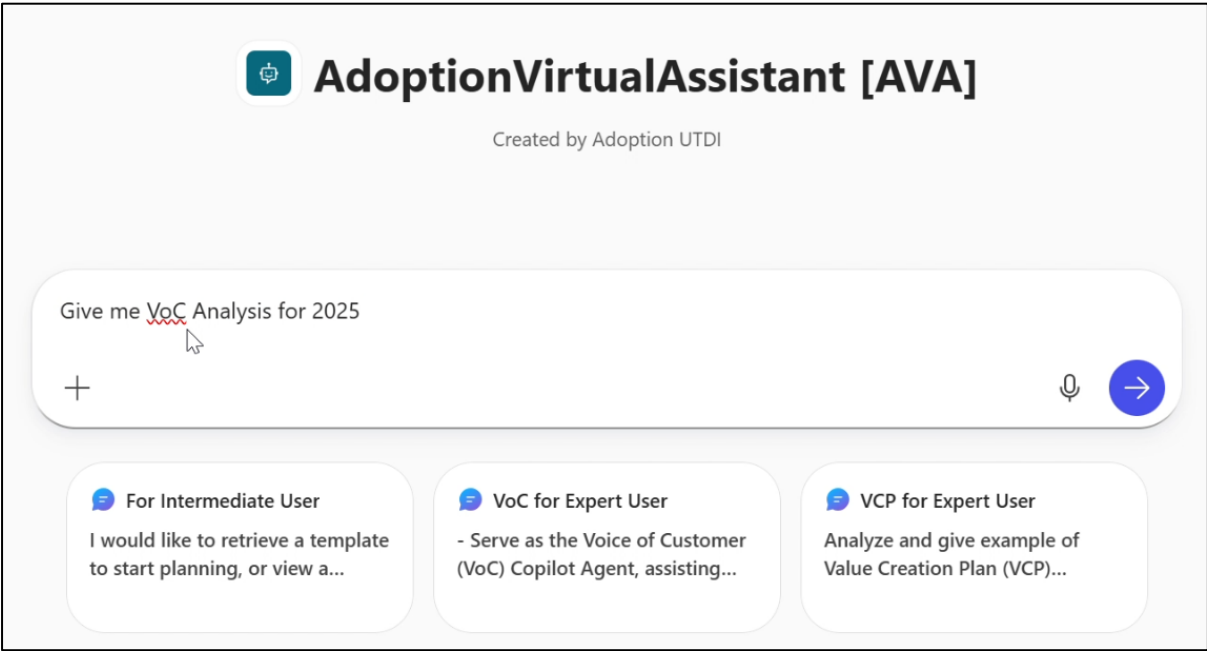


Figure 33: AVA Matured Agent : User asking a Matured-level question

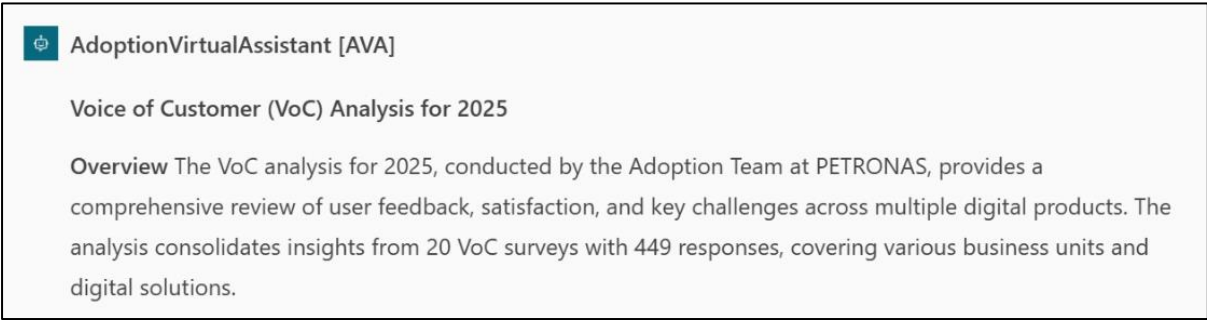


Figure 34: AVA Matured Agent Output (i)

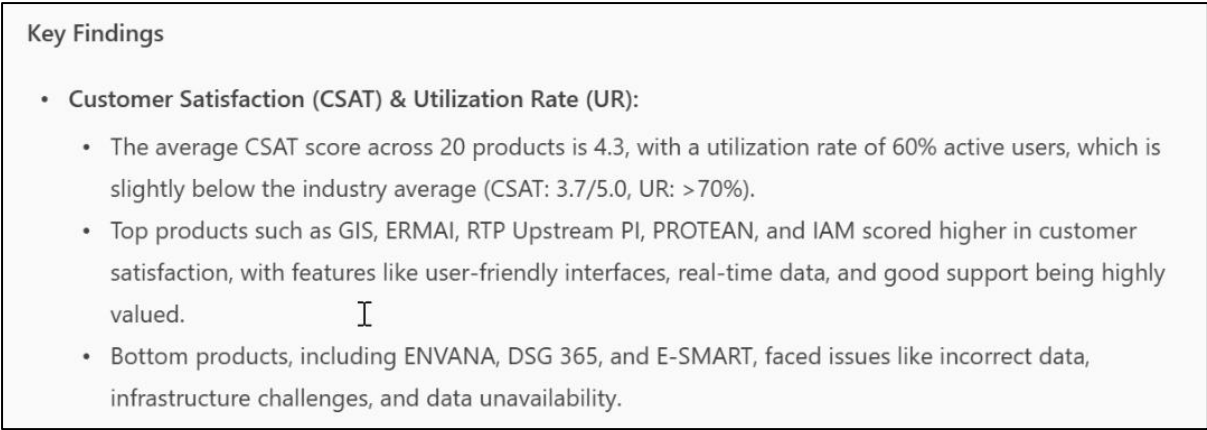


Figure 35: AVA Matured Agent Output (ii)

• **Major Issues Identified:**

- **Integration and Cost:** Integration between systems remains problematic, and the cost of implementation is often seen as outweighing the benefits. There is also a lack of a continuous team to drive utilization and improvement.
- **Data Quality & ICT Requirements:** Discrepancies between service provider data and in-house tracking, as well as the need for stable and dedicated internet bandwidth, are recurring concerns.
- **Technical Support & User Adaptation:** Delays in technical support due to time zone differences and the need for users to quickly adapt to new workflows are significant challenges. Users sometimes revert to manual control for safety reasons.
- **User Experience & Interface:** Users report confusing and hard-to-navigate interfaces, slow performance, and lack of engaging UI/UX. Workshops and training are recommended to address these issues.

Figure 36: AVA Matured Agent Output (iii)

Action Plans & Recommendations

- Conduct UI/UX workshops with key users to improve adaptability and site navigation.
- Align product owners with Wells Management to ensure consistent usage and provide refresher training for all users.
- Enhance technical support processes, especially for users in different time zones.
- Collaborate with data teams to ensure relevant and accurate data is available for daily operations.
- Address ICT and connectivity issues to improve accessibility and application performance.
- Implement a gap closure plan focusing on integration, cost-benefit analysis, and continuous improvement teams. VoC Analysis 2025.pptx +2

Figure 37: AVA Matured Agent Output (iv)

Product-Specific Insights

- **FPC Product:** Achieved a 77% usage rate and a CSAT score of 3.63, slightly below the industry average. Major issues include data quality, ICT infrastructure, restricted access to modules, and non-intuitive customization features. Users also noted the need for consistent data updates and better visualization options. VoC Analysis 2025.pptx

Summary The 2025 VoC analysis highlights both strengths and areas for improvement across PETRONAS digital products. While some products are well-received for their intuitive interfaces and support, others require significant enhancements in integration, data quality, user training, and technical support to elevate overall customer experience and satisfaction. VoC Analysis 2025.pptx +2

: cite:1 "Citation-1": cite:2 "Citation-2": cite:3 "Citation-3" VoC Analysis 2025.pptx VoC Analysis 2025.pptx VoC Analysis 2025.pptx

Figure 38: AVA Matured Agent Output (v)

During the testing of the AVA Matured Agent, the user submitted a query that required a high level of expertise: “Give me VoC Analysis for 2025.” Recognising the advanced nature of the question, AVA automatically activated its Matured Agent mode to ensure a response that matched the complexity and depth expected at this level. The agent then examined the relevant knowledge documents, adhering closely to the instructions and logic embedded within its topic configurations.

Through this analytical process, AVA was able to generate a comprehensive summary of the VoC Analysis for 2025. By scanning the contents of the VoC Analysis document, the agent accurately extracted key insights and produced a summary that was both precise and well-aligned with the original document. This test highlighted the Matured Agent’s capability to interpret complex queries, synthesise detailed information from multiple sources, and deliver high-quality, expert-level responses that are directly applicable to strategic decision-making in digital technology adoption.

6.3 Adoption Virtual Assistant (AVA) Performance Post-Deployment

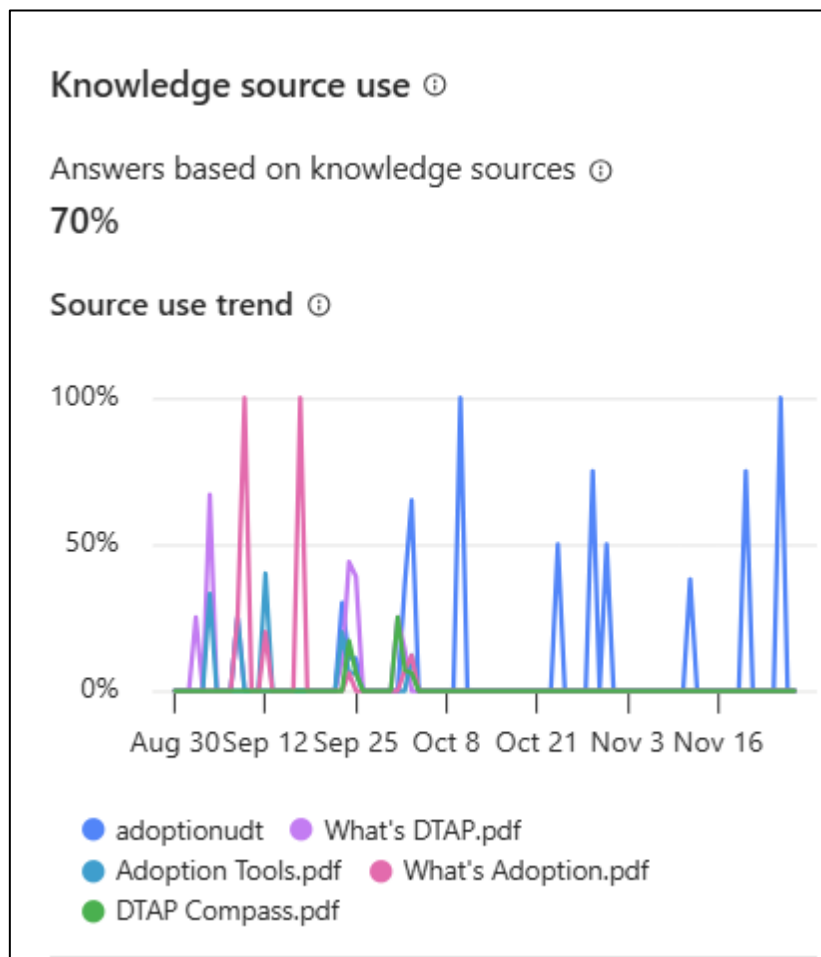


Figure 39: AVA Analytics : Knowledge Source Use

According to the diagram, AVA achieved a positive score of 70% when using its uploaded sources to respond to user inquiries. This metric is particularly important, as maintaining accuracy is crucial for AVA to effectively support UTDI users in their Digital Technology

Adoption journey. A lower score would suggest a higher likelihood of hallucinated or incorrect responses, which could undermine the system’s reliability and its ability to provide useful, actionable guidance. Achieving a high accuracy rate ensures that users can trust AVA’s outputs and rely on the system for informed decision-making and practical adoption strategies.

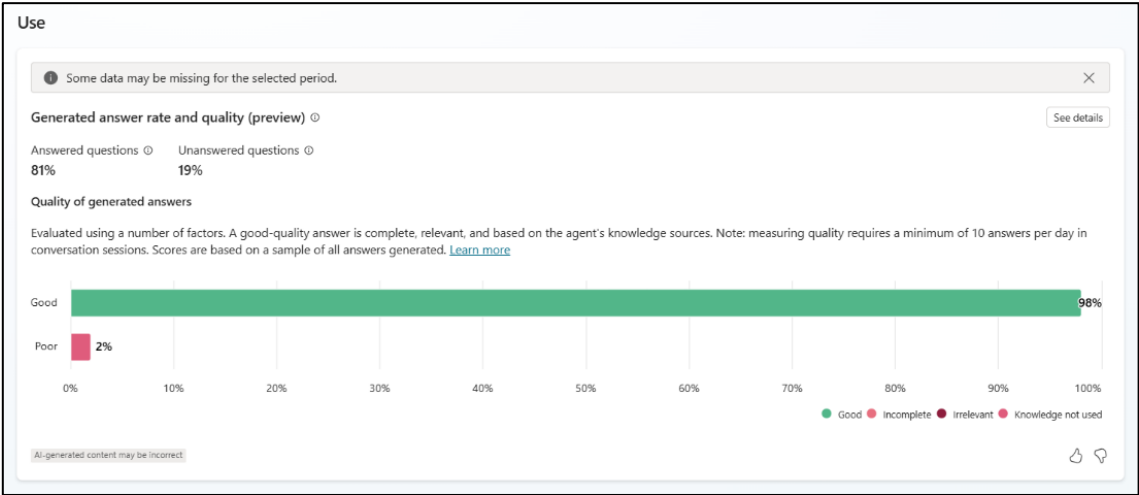


Figure 40: AVA Analytics : General Answer Rate and Quality

Based on the diagram, AVA achieved a rating of 81% for Answered Questions and 19% for Unanswered Questions. This positive performance demonstrates AVA’s ability to recognise user inquiries, analyse its knowledge sources, and generate carefully curated responses. Overall, the high percentage of answered questions indicates that AVA can successfully identify and respond to the majority of user queries. The 19% of unanswered questions could be attributed to factors such as the user asking questions outside AVA’s defined scope, submitting irrelevant queries, or including typographical errors.

In addition, AVA attained a remarkable 98% rating for the Quality of Generated Answers. This metric highlights that the responses provided by AVA are largely accurate, complete, and relevant, aligning closely with the information contained in its uploaded knowledge sources. The high quality score underscores AVA’s capability to deliver reliable and meaningful outputs, reinforcing its effectiveness as a tool for supporting users in Digital Technology Adoption.

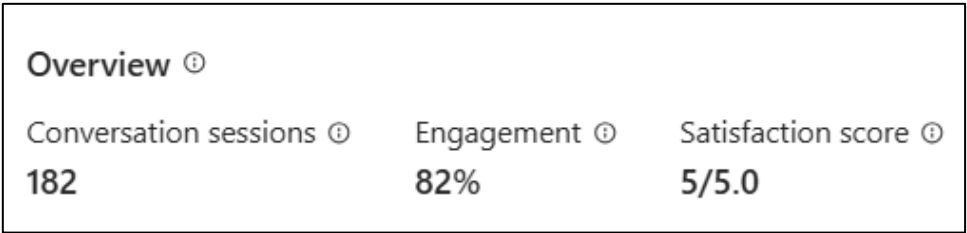


Figure 41: AVA Analytics : Conversation, Engagement, User Satisfaction Overview

The diagram provides an overview of AVA's performance in terms of Conversation Sessions, User Engagement, and User Satisfaction Score. Since its deployment, AVA has recorded a substantial total of 182 Conversation Sessions, reflecting frequent interactions with users. In addition, AVA achieved a User Engagement rating of 81% and an impressive User Satisfaction Score of 5 out of 5.

These metrics indicate that AVA has been highly effective in maintaining active engagement among UTDI personnel. High User Engagement demonstrates that users interact with AVA repeatedly, suggesting that the tool is valued and consistently utilised by staff for their digital technology adoption needs. Furthermore, the perfect User Satisfaction Score highlights that users are consistently pleased with AVA's performance across all 182 Conversation Sessions. Overall, these results provide strong evidence that AVA not only facilitates meaningful interactions but also delivers a reliable and satisfactory user experience throughout its deployment.

6.4 Adoption Virtual Assistant (AVA) Sustainability

Sustainability Aspects of the AVA Project

The Adoption Virtual Assistant (AVA), developed as an AI-powered solution using Microsoft Copilot Studio, contributes to sustainability within PETRONAS's Upstream Technology Digital Innovation (UTDI) department by facilitating the efficient adoption of digital technologies. By automating guidance and support for digital adoption processes, AVA promotes operational efficiency, workforce empowerment, and economic benefits, reflecting broader trends in which AI supports sustainable development. The sustainability impact of AVA can be examined through economic growth, industrial innovation, and workforce development, aligning closely with the United Nations Sustainable Development Goals (SDGs), specifically SDG 8 and SDG 9.

Economic Sustainability and SDG 8 : Decent Work and Economic Growth

From an economic perspective, AVA enhances sustainability by streamlining the adoption of digital technologies, reducing operational costs, and improving overall efficiency. By replacing the traditional Adoption Team with an automated, always-available virtual assistant, AVA reduces human resource dependencies, lowers training and support expenses, and accelerates the implementation of cost-saving AI applications, such as predictive maintenance, supply chain optimization, and drilling operations. These improvements increase productivity, reduce downtime, and strengthen financial resilience, supporting SDG 8: Promote sustained, inclusive, and sustainable economic growth, full and productive employment, and decent work for all. Furthermore, AVA accommodates users with varying expertise levels, ensuring efficient knowledge transfer, fostering innovation, and generating long-term economic value by enabling scalable adoption of technologies that enhance both profitability and workforce development.

Industrial Innovation and SDG 9 : Industry, Innovation, and Infrastructure

In terms of industrial sustainability, AVA contributes directly to SDG 9: Build resilient infrastructure, promote inclusive and sustainable industrialization, and foster innovation by facilitating the adoption of advanced AI tools that improve operational efficiency, reduce resource consumption, and minimize environmental risks during drilling and exploration activities. By integrating digital technologies seamlessly into PETRONAS's operations, AVA strengthens technological capabilities, drives modernization within the oil and gas sector, and promotes sustainable industrialization and innovation across upstream activities.

Social and Workforce Empowerment

AVA also advances social sustainability by empowering employees across all levels of expertise to effectively engage with digital tools. By automating routine adoption tasks and providing accessible guidance, AVA promotes AI literacy and skill development, enabling employees to focus on higher-value tasks and strategic decision-making. This supports smoother organizational transitions, mitigates risks of job displacement, and fosters safer working environments through enhanced monitoring and adherence to ESG standards. By promoting broader AI adoption, AVA cultivates an inclusive and innovative culture that balances technological advancement with workforce well-being.

7.0 Conclusion and Recommendation

In conclusion, the development of the Adoption Virtual Assistant (AVA) has successfully fulfilled its primary objective of establishing a platform that provides accessible, reliable, and accurate knowledge on adoption-related matters. Using Microsoft Copilot Studio, a personalized AI Agent was created to guide users through the digital technology adoption journey, encouraging independent engagement with established adoption tools and methodologies. The Adoption Agent delivers real-time support, contextual insights, and tailored recommendations, effectively replicating the guidance that would normally be provided by adoption personnel. This capability has empowered digital product managers and project managers to take full ownership of adoption initiatives, allowing them to plan, execute, and monitor digital adoption efforts independently in accordance with the objectives of the Adoption Franchise Program (AFP).

The three child agents within AVA, namely the Beginner, Intermediate, and Matured Agents, have demonstrated their effectiveness in providing structured guidance tailored to different levels of user expertise. Each agent addresses the unique needs of its respective audience, ensuring that the information and support offered are relevant and appropriate for the user's level of knowledge. Analytics results indicate that AVA is successfully facilitating user engagement and adoption processes, establishing itself as a valuable tool that contributes significantly to the operational and strategic goals of PETRONAS Upstream. These results suggest that AVA is on track to become a widely used resource that supports digital transformation initiatives within the Upstream Technology Digital Innovation department.

Looking forward, there are several opportunities to enhance AVA's capabilities and expand its impact. One recommendation is to broaden the scope of AVA beyond digital technology adoption matters. By incorporating functionalities such as organizational planning, performance tracking, digital product value monitoring, and departmental financial management, AVA could evolve into a comprehensive digital assistant that supports strategic decision-making across multiple areas of UTDI operations.

Additionally, deploying AI Agents in other teams and departments within UTDI could replicate the success achieved by this project, promoting consistent adoption practices and wider utilization of digital tools across the organization. Another suggested enhancement is to expand the persona selection options beyond the current Beginner, Intermediate, and Matured levels. Introducing role-based personas, such as Executive, Manager, and Senior Manager, would allow AVA to better understand user context, responsibilities, and priorities, enabling it to provide more relevant, actionable, and personalized recommendations.

Implementing these improvements will ensure that AVA continues to evolve as a robust and scalable solution, capable of addressing complex organizational needs while supporting PETRONAS' broader digital transformation and sustainability goals. By combining personalized guidance, advanced analytics, and an expanded operational scope, AVA has the potential to become an indispensable tool that enhances productivity, fosters innovation, and strengthens the overall efficiency and effectiveness of digital technology adoption across the Upstream Technology Digital Innovation department.

8.0 References and Citation

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